

claude-windows / 9f6357d8-0e35-491a-a830-052e094c7f37

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_b7edcaa8-69d\agent-ac3e67dafdc3abf3.jsonl`
- SHA-256: `9422eeb2d767f0ff41a3601510fa838673cc22b45392a74464d1afd9cd1f1f3b`
- Source modified: `2026-06-21T14:37:01+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_b7edcaa8-69d`
- session_id: `9f6357d8-0e35-491a-a830-052e094c7f37`
```

Transcript

1. user (2026-06-21T14:33:51.168Z)

Spec the **B3 ticket** (carry the trilingual win to the marketing site). Read in C:/Users/avidu/Projects/khelsutra: site/public/{index,capture,own-model}.html, site/server.js, site/src/assets.test.js, web/src/i18n/index.ts + web/src/i18n/locales/en/common.json (the catalog format + negotiation), and docs/FRIENDLY_AND_MULTILINGUAL.md §4/§7-B3 + the engine docs/I18N_PLAN.md P1 (the vanilla t() shim design). Produce ONE ticket (id "B3") with: the tiny t(key,vars)/data-i18n shim design (where it lives, ~LOC, how it walks data-i18n attributes + mirrors i18next interpolation/2-form plurals), the **catalog-home decision** (hoist web/src/i18n/locales to a shared top-level locales/ consumed by BOTH surfaces, vs the site importing from web/src/i18n/locales ? recommend one, with the build/bundling implication for site/), the negotiation parity (persisted pref ? ?lang= ? Accept-Language ? en, matching the SPA), the Accept-Language server test in site/, and the test plan. Return structured JSON (tickets:[B3]).

2. assistant (2026-06-21T14:34:00.924Z)

I'll spec the B3 ticket. Let me start by pulling the khelsutra repo fresh and reading all the referenced files.

3. user (2026-06-21T14:34:04.809Z)

Already up to date.

```
---BRANCH---
```

```
master
```

```
---HEAD---
```

```
a20002f docs: propose Multilingual Expansion ? 7 new locales (th/id/da/es/zh-Hans/ja/ko) (#58)
```

4. user (2026-06-21T14:34:04.819Z)

```
1      // Always-on VM host for the KhelSutra site ? dependency-free (Node 22+ only).
2      // Serves public/ and POST /api/request-access into SQLite (node:sqlite), using the
3      // SAME handler logic as the Cloudflare Worker (src/index.js). This is the portability
4      // fallback "if Cloudflare acts out" ? see docs/PORTABILITY.md.
5      //
6      //   PORT=8080 DB_PATH=/var/lib/khelsutra/access.db node server.js
7      //
8      import { createServer } from "node:http";
9      import { readFile } from "node:fs/promises";
10     import { join, extname, normalize } from "node:path";
11     import { DatabaseSync } from "node:sqlite";
12     import { handleRequest } from "./src/index.js";
13     import { sqliteStore } from "./src/store.js";
14     import { noopNotify } from "./src/notify.js";
15
16     const PORT = Number(process.env.PORT) || 8080;
```

```

17     const ROOT = import.meta.dirname;
18     const PUBLIC = join(ROOT, "public");
19     const DB_PATH = process.env.DB_PATH || join(ROOT, "access.db");
20
21     const db = new DatabaseSync(DB_PATH);
22     db.exec(await readFile(join(ROOT, "schema.sql"), "utf8")); // CREATE TABLE IF NOT EXISTS
? idempotent
23     const store = sqliteStore(db);
24     const notify = noopNotify(); // VM host: no email yet (add SMTP later ? see
docs/PORTABILITY.md)
25
26     const TYPES = {
27         ".html": "text/html; charset=utf-8", ".png": "image/png", ".svg": "image/svg+xml",
28         ".css": "text/css", ".js": "text/javascript", ".ico": "image/x-icon", ".txt":
"text/plain",
29         ".mp4": "video/mp4", ".webm": "video/webm", ".gif": "image/gif", ".jpg": "image/jpeg",
".jpeg": "image/jpeg",
30     };
31
32     async function toRequest(req) {
33         const url = `http://${req.headers.host || "localhost"}${req.url}`;
34         let body;
35         if (req.method !== "GET" && req.method !== "HEAD") {
36             const chunks = [];
37             for await (const c of req) chunks.push(c);
38             body = Buffer.concat(chunks);
39         }
40         return new Request(url, { method: req.method, headers: req.headers, body });
41     }
42
43     async function send(res, response) {
44         res.statusCode = response.status;
45         for (const [k, v] of response.headers) res.setHeader(k, v);
46         res.end(Buffer.from(await response.arrayBuffer()));
47     }
48
49     async function serveStatic(pathname, res) {
50         const rel = pathname === "/" ? "index.html"
51             : pathname === "/capture" ? "capture.html"
52             : pathname === "/own-model" ? "own-model.html"
53             : pathname === "/flyer" ? "flyer.html"
54             : pathname === "/flyer-kn" ? "flyer-kn.html"
55             : pathname === "/flyer-hi" ? "flyer-hi.html"
56             : pathname.replace(/^\/+/, "");
57         const file = normalize(join(PUBLIC, rel));
58         if (!file.startsWith(PUBLIC)) { res.statusCode = 403; return res.end("forbidden"); }
59         try {
60             const data = await readFile(file);
61             res.setHeader("content-type", TYPES[extname(file)] || "application/octet-stream");
62             res.end(data);
63         } catch {
64             res.statusCode = 404;
65             res.end("not found");
66         }
67     }
68
69     createServer(async (req, res) => {
70         try {
71             const pathname = decodeURIComponent(new URL(`http://${req.url}`).pathname);
72             if (pathname === "/api/request-access") {
73                 if (req.method !== "POST") { res.statusCode = 405; return
res.end('{"ok":false,"error":"method not allowed"}'); }
74                 return await send(res, await handleRequest(await toRequest(req), store, notify));
75             }
76             return await serveStatic(pathname, res);

```

```

77     } catch (err) {
78         res.statusCode = 500;
79         res.end({'ok':false,"error":"server error"});
80         console.error(err);
81     }
82     }).listen(PORT, () => console.log(`khelsutra site listening on :${PORT} (sqlite:
${DB_PATH}`));
83

```

5. user (2026-06-21T14:34:04.820Z)

```

1      // Static-site integrity ? the landing page + capture checklist + OG image ARE the
public
2      // product. server.test.js proves the VM host *serves* "/", "/capture", "/og.png"; this
3      // guards the CONTENT contract so a broken edit (deleted asset, broken form wiring, dead
4      // capture link, missing/!absolute OG image) fails CI instead of shipping to the live
site.
5      import { test } from "node:test";
6      import assert from "node:assert/strict";
7      import { readFileSync, statSync } from "node:fs";
8      import { join } from "node:path";
9
10     const PUBLIC = join(import.meta.dirname, "..", "public");
11     const read = (f) => readFileSync(join(PUBLIC, f), "utf8");
12     const size = (f) => statSync(join(PUBLIC, f)).size;
13
14     test("core public assets exist and are non-empty", () => {
15         for (const f of ["index.html", "capture.html", "og.png"]) {
16             assert.ok(size(f) > 0, `public/${f} is missing or empty`);
17         }
18     });
19
20     test("landing page wires the Request-Access form to the API", () => {
21         const html = read("index.html");
22         assert.match(html, /id="af"/, "the #af form is gone");
23         assert.match(html, /\api\/request-access/, "form no longer posts to /api/request-
access");
24         assert.match(html, /name="company"[\^>]*tabindex="-1"/, "honeypot field removed (bots
will get through)");
25         for (const field of ["segment", "has_gpu", "storage"]) {
26             assert.match(html, new RegExp(`name="${field}"`), `form lost the '${field}' field (a
demand signal)`);
27         }
28     });
29
30     test("submitting the form shows an obvious in-flight state (no silent freeze)", () => {
31         const html = read("index.html");
32         assert.match(html, /Sending?/, "submit should swap to a 'Sending?' state while the
request is in flight");
33         assert.match(html, /aria-busy/, "submit should set aria-busy for assistive tech during
the request");
34         assert.match(html, /class="spin"/, "the in-flight spinner is gone");
35         assert.match(html, /\spin\{/, "the spinner style is gone");
36         // the button must be RE-ENABLED on failure so the user isn't stuck on a frozen
'Sending?'
37         assert.match(html, /btn\.disabled\s*=\s*false/, "submit button isn't restored on the
failure path");
38     });
39
40     test("OG + canonical meta point at the live https origin", () => {
41         const html = read("index.html");
42         assert.match(html,
/property="og:image"\s+content="https:\/\/khelsutra.guru\/og.png"/, "og:image must be the
absolute live URL");
43         assert.match(html, /property="og:url"\s+content="https:\/\/khelsutra.guru"/, "og:url

```

```

drift");
44     });
45
46     test("the sample reel is GATED behind sign-up + starts at a RANDOM point (not shown
openly)", () => {
47         const html = read("index.html");
48         assert.match(html, /ajTCZ7pJW_o/, "the sample-reel video id is gone");
49         assert.match(html, /youtube-nocookie\.com/, "reel should use the privacy nocookie
host");
50         assert.match(html, /id="reel-reveal"/, "the on-signup reveal host is gone");
51         assert.match(html, /see it in action/i, "the 'see it in action' sign-up teaser is
gone");
52         assert.match(html, /reel-count/, "the auto-close countdown is gone");
53         assert.match(html, /REEL_SECS = 30/, "the reel preview should be ~30s");
54         assert.match(html, /getDuration\(\)/, "random start should derive from the real video
duration");
55         assert.match(html, /Math\.random\(\)/, "the start point should be randomized");
56         assert.match(html, /seekTo\(/, "the player should seek to the random start");
57         // Must NOT be an openly-rendered public section ? it only appears after a successful
submit.
58         assert.doesNotMatch(html, /<section id="reel"/, "the reel must be sign-up-gated, not
an open public section");
59         assert.doesNotMatch(html, /id="reel-play"/, "the open click-to-play facade should be
removed");
60     });
61
62     test("the TOOL screencast is self-hosted, footage-free, and revealed only after sign-
up", () => {
63         const html = read("index.html");
64         assert.match(html, /\video\tool-cuj\.mp4/, "the tool-operation screencast reference
is gone");
65         assert.match(html, /function revealAfterSignup/, "the post-signup reveal function is
gone");
66         assert.match(html, /revealAfterSignup\(\)/, "the reveal must be wired to a successful
submit");
67         // GATED: the only reference to the tool screencast lives inside the reveal function
(injected on
68         // signup), not as an open autoplay element rendered for every visitor.
69         assert.ok(
70             html.indexOf("/video/tool-cuj.mp4") > html.indexOf("function revealAfterSignup"),
71             "the tool screencast must be inside the post-signup reveal, not shown openly",
72         );
73         // the self-hosted clip exists and is non-trivial (a real recording, not an empty
placeholder)
74         assert.ok(size("video/tool-cuj.mp4") > 50_000, "tool-cuj.mp4 is missing or too small
to be the real screencast");
75     });
76
77     test("the post-signup reveal keeps its watchability affordances (manual close + play-
gated auto-close)", () => {
78         // Content contract only ? the BEHAVIOUR (close works, auto-close holds while the reel
plays) is
79         // proven against the real page in ux.dom.test.js, so we don't pin brittle
implementation strings.
80         const html = read("index.html");
81         assert.match(html, /id="reel-close"/, "the manual ? close button is gone");
82         assert.match(html, /onStateChange/, "the reel's play-state must still gate the auto-
close");
83     });
84
85     test("the capture checklist is linked and is a real page", () => {
86         assert.match(read("index.html"), /href="\capture\(.html\)?"/, "footer capture-
checklist link is gone");
87         const cap = read("capture.html");
88         assert.match(cap, /<html/i, "capture.html is not HTML");

```

```

89     assert.match(cap, /capture checklist/i, "capture.html lost its checklist content");
90 });
91
92 test("the academy flyers (en/kn/hi) exist, are language-correct, and are linked from the
homepage", () => {
93     const html = read("index.html");
94     // All three are linked from the homepage footer (the owner ask: the flyers must be
findable).
95     assert.match(html, /href="\ flyer"/, "homepage doesn't link the English academy
flyer");
96     assert.match(html, /href="\ flyer-kn"/, "homepage doesn't link the Kannada academy
flyer");
97     assert.match(html, /href="\ flyer-hi"/, "homepage doesn't link the Hindi academy
flyer");
98
99     // The three flyer pages exist, are HTML, and carry the right language + script.
100    const en = read("flyer.html");
101    assert.match(en, /<html lang="en"/i, "flyer.html should be lang=en");
102    const kn = read("flyer-kn.html");
103    assert.match(kn, /<html lang="kn"/i, "flyer-kn.html should be lang=kn");
104    assert.match(kn, /[? -?]/, "flyer-kn.html should contain Kannada text");
105    const hi = read("flyer-hi.html");
106    assert.match(hi, /<html lang="hi"/i, "flyer-hi.html should be lang=hi");
107    assert.match(hi, /[? -?]/, "flyer-hi.html should contain Devanagari (Hindi) text");
108 });
109
110 test("each flyer shows a screenshot of the tool rendered in that language", () => {
111     for (const [flyer, img] of [
["flyer.html", "studio-en.png"],
["flyer-kn.html", "studio-kn.png"],
["flyer-hi.html", "studio-hi.png"]
]) {
112         assert.ok(size(`img/${img}`) > 0, `public/img/${img} (the localized UI shot) is
missing or empty`);
113         assert.match(read(flyer), new RegExp(`img/${img}`), `${flyer} should embed its
localized UI screenshot /img/${img}`);
114     }
115 });
116
117 test("the academy flyers are also linked from the top nav and the own-model page", () =>
{
118     const html = read("index.html");
119     // Desktop top nav carries a Flyer entry (alongside Own your model).
120     const nav = html.match(/<nav class="nav-links"[s\S]*?<\/nav>/);
121     assert.ok(nav, "the primary nav block is gone");
122     assert.match(nav[0], /href="\ flyer"/, "the top nav doesn't link the Flyer");
123     // own-model surfaces all three languages.
124     const om = read("own-model.html");
125     assert.match(om, /href="\ flyer"/, "own-model doesn't link the English flyer");
126     assert.match(om, /href="\ flyer-kn"/, "own-model doesn't link the Kannada flyer");
127     assert.match(om, /href="\ flyer-hi"/, "own-model doesn't link the Hindi flyer");
128 });
129
130 test("the landing page surfaces how to record reliably (so users can set up with
confidence)", () => {
131     const html = read("index.html");
132     assert.match(html, /id="record"/, "the 'record it right' section is gone");
133     assert.match(html, /one match, one court/i, "the #1 single-court capture rule should
be on the homepage");
134     // the gist must carry the load-bearing capture cues + a prominent link to the full
checklist
135     for (const cue of [
"/1080p/",
"/30 fps/",
"/[Aa]udio ON/",
"/baseline/",
"/[Nn]o panning/
]) {
136         assert.match(html, cue, `the recording gist is missing a key cue: ${cue}`);
137     }
138     assert.match(html, /href="\ capture"[^>]*>Full 1-page capture checklist/i, "a
prominent link to the full checklist is gone");
139 });
140

```

```

141     test("the 'How do I join?' FAQ points at the request-access form, not email", () => {
142         const html = read("index.html");
143         const m = html.match(/How do I join\?<\summary>(<p[\s\S]*?<\p>)/);
144         assert.ok(m, "the 'How do I join?' FAQ is gone");
145         assert.match(m[1], /request access/i, "the join FAQ should send people to the request-
access form");
146         assert.match(m[1], /href="#join"/, "the join FAQ should link to the #join form");
147         assert.doesNotMatch(m[1], /mailto/, "the join FAQ should NOT tell people to email (use
the form)");
148     });
149
150     test("the own-model page exists, is honestly scoped, and is linked from the landing
page", () => {
151         const om = read("own-model.html");
152         assert.ok(om.length > 0, "own-model.html is missing or empty");
153         for (const id of ["annotate", "train", "setup"]) {
154             assert.match(om, new RegExp(`id="${id}"`), `own-model is missing the #${id}
section`);
155         }
156         assert.match(om, /github\.com\/Khelsutra\/rally-annotator/, "the GitHub link to the
annotator is gone");
157         assert.match(om, /Read this honestly/i, "the honest framing fence is gone");
158         assert.match(om, /open pipeline|in build/i, "training must be honestly scoped as in-
build, not one-click");
159         assert.match(om, /name="robots" content="noindex"/, "own-model should be noindex");
160         assert.match(read("index.html"), /href="\own-model"/, "the landing page should link
to /own-model");
161         assert.match(om, /<a href="\/">? KhelSutra home<\a>/, "own-model needs an obvious
back-to-home link (not just browser-back)");
162
163         // a ready-made default works without training (keeps non-tech users engaged); BYO-
training is optional
164         assert.match(om, /ready-made default|out of the box|don't have to train/i, "must
reassure a default works without training");
165         assert.match(om, /optional/i, "training your own should be framed as optional");
166
167         // own-model shows the annotation TOOL screenshots (no people) ? real captures, files
present
168         for (const img of ["annotate-enable.png", "annotate-tool.png"]) {
169             assert.match(om, new RegExp(`/img/${img}`), `own-model should embed /img/${img}`);
170             assert.ok(size(`img/${img}`) > 0, `public/img/${img} is missing or empty`);
171         }
172         assert.match(om, /<img[^>]+alt="/i, "screenshots must have alt text");
173
174         // the sample REEL stays gated behind request-access ? NOT embedded/playable here, no
footage shown
175         assert.doesNotMatch(om, /<iframe/i, "the reel must NOT be embedded here ? it's gated
behind request-access");
176         assert.doesNotMatch(om, /youtube/i, "no YouTube embed on own-model; the reel is shown
only after request-access");
177         assert.doesNotMatch(om, /i\.yimg\.com/, "don't show the video thumbnail (players)
openly");
178         assert.match(om, /sample[\s\S]{0,80}request access|request access[\s\S]{0,80}sample/i,
"point users to request access for a sample output");
179     });
180

```

6. assistant (2026-06-21T14:34:06.950Z)

Let me read the remaining files in parallel.

7. user (2026-06-21T14:34:07.884Z)

```

1     // i18n scaffold (tracker B1) ? the shared localization foundation for KhelSutra Studio.
2     //

```

```

3      // Per docs/FRIENDLY_AND_MULTILINGUAL.md + the engine's I18N_PLAN.md:
4      //   - keys, never prose (t("app.title"), enforced later by a no-bare-strings lint, B2);
5      //   - ICU MessageFormat for interpolation/plurals (Kannada/Hindi plural rules live in
the catalog,
6      //     not the code ? replaces the SPA's hand-rolled `count > 1 ? "s" : ""`);
7      //   - one negotiation order shared with the marketing site (B3);
8      //   - always-fallback to `en` so a missing translation degrades to English, never a raw
key.
9      import i18n from "i18next";
10     import { initReactI18next } from "react-i18next";
11     import LanguageDetector from "i18next-browser-languagedetector";
12     import ICU from "i18next-icu";
13     import enCommon from "../locales/en/common.json";
14     import knCommon from "../locales/kn/common.json";
15     import hiCommon from "../locales/hi/common.json";
16
17     // Supported locales (D6): `en` + `kn` + `hi`. Anything else falls back to `en`.
18     export const SUPPORTED_LOCALES = ["en", "kn", "hi"] as const;
19     export type Locale = (typeof SUPPORTED_LOCALES)[number];
20     export const DEFAULT_LOCALE: Locale = "en";
21
22     // ONE persisted-preference key, shared with the marketing site's t() shim (D9 / tracker
B3), so a
23     // visitor's language choice carries across both surfaces.
24     export const LANG_STORAGE_KEY = "khelsutra.lang";
25
26     // Catalogs are co-located here (D9). `kn` + `hi` shipped (B5/B6, AI-drafted ? native
review
27     // pending). A missing key in any locale still falls back to `en` ? the by-design
invariant.
28     const resources = {
29       en: { common: enCommon },
30       kn: { common: knCommon },
31       hi: { common: hiCommon },
32     } as const;
33
34     void i18n
35       .use(ICU)
36       .use(LanguageDetector)
37       .use(initReactI18next)
38       .init({
39         resources,
40         fallbackLng: DEFAULT_LOCALE,
41         supportedLngs: [...SUPPORTED_LOCALES],
42         nonExplicitSupportedLngs: true, // en-IN -> en
43         defaultNS: "common",
44         ns: ["common"],
45         interpolation: { escapeValue: false }, // React already escapes
46         returnNull: false,
47         // Negotiation order (I18N_PLAN §2.3): persisted pref -> ?lang= -> browser -> en.
48         detection: {
49           order: ["localStorage", "querystring", "navigator"],
50           lookupLocalStorage: LANG_STORAGE_KEY,
51           lookupQuerystring: "lang",
52           caches: ["localStorage"],
53         },
54         react: { useSuspense: false },
55       });
56
57     // Keep <html lang> in sync with the active locale (screen readers + SEO use it).
Reflects the
58     // resolved content language, so an `en` fallback for an absent locale still reads as
`en`.
59     function syncDocumentLang(lng: string) {
60       if (typeof document !== "undefined") {

```

```

61     document.documentElement.lang = (i18n.resolvedLanguage || lng ||
DEFAULT_LOCALE).split("-")[0];
62     }
63     }
64     i18n.on("languageChanged", syncDocumentLang);
65     syncDocumentLang(i18n.language ?? DEFAULT_LOCALE);
66
67     export default i18n;
68

```

8. user (2026-06-21T14:34:08.377Z)

```

1      {
2      "app": {
3          "badge": "per-rally selection · alpha",
4          "title": "Pick the rallies for your reel",
5          "intro": "Filter by <b>rally length</b> ? e.g. <i>?top 15 longest?</i> ? then fine-
tune by tapping cards. Everything starts <b>selected</b>; just drop the ones you don't want."
6      },
7      "lang": {
8          "label": "Language",
9          "en": "English",
10         "kn": "?????",
11         "hi": "?????"
12     },
13     "bulk": {
14         "selectAll": "Select all",
15         "clear": "Clear",
16         "invert": "Invert"
17     },
18     "mode": {
19         "demo": "Demo data ? open with <code>?video=?video_id?</code> against a running
engine to load a real session.",
20         "loading": "Loading rallies for <code>{videoId}</code>?",
21         "error": "Couldn't load rallies for <code>{videoId}</code>: {error}",
22         "loaded": "Loaded <b>{count}</b> rallies from this session."
23     },
24     "query": {
25         "placeholder": "e.g. over 10s · top 15 longest · shortest first",
26         "ariaLabel": "Filter rallies by length, count or order",
27         "apply": "Apply",
28         "notAvailable": "not available yet",
29         "notAvailableTitle": "?{token}? needs {kind} detection ? not available yet",
30         "unreadable": "Couldn't read that ? try ?over 10s? or ?top 15 longest?.",
31         "hint": "Type over 10s for rallies longer than 10 seconds ? or tap a sample below.",
32         "try": "Sample queries:",
33         "support": "Filtering by <b>rally length</b> is fully supported. Shot-count is
<x>experimental</x> (under development)."
34     },
35     "grid": {
36         "empty": "No rallies to show yet.",
37         "rally": "Rally {n}",
38         "inReel": "Rally {n}, in reel",
39         "notInReel": "Rally {n}, not in reel",
40         "start": "start",
41         "long": "long",
42         "shots": "~{count} shots · experimental",
43         "shotsTitle": "Shot count is experimental (under development) and may be
inaccurate{conf}. Filter by rally length instead.",
44         "shotsConfidence": " ? confidence: {conf}"
45     },
46     "footer": {
47         "count": "{count, plural, one {# rally} other {# rallies}}",
48         "summaryRest": "selected · {total} total",
49         "reelName": "Reel name",

```

```

50     "reelNamePlaceholder": "my-highlights",
51     "build": "Build reel ({count})",
52     "building": "Building?",
53     "floorOne": "<b>Your only selected rally</b> has fewer than {min} shots ? the engine
would skip it, so the build would fail. Pick a longer rally.",
54     "floorAll": "<b>All {count} selected rallies</b> have fewer than {min} shots ? the
engine skips every one, so the build would fail. Pick longer rallies.",
55     "floorSomeLead": "{count, plural, one {# rally} other {# rallies}}",
56     "floorSomeRest": "with fewer than {min} shots will be skipped when building (the
engine's <code>min_shot_count</code> floor).",
57     },
58     "result": {
59         "ready": "Your reel is ready",
60         "failed": "Build failed",
61         "dismiss": "Dismiss",
62         "download": "Download {filename}"
63     },
64     "build": {
65         "noVideo": "Open with ?video=<video_id> against a running engine to build a reel ?
demo data can't be compiled."
66     },
67     "ingest": {
68         "title": "Process a video",
69         "intro": "Point KhelSutra at a video ? a bucket link or a file on the engine ? and
it finds the rallies. This can take a few minutes.",
70         "uriPlaceholder": "gs://your-bucket/match.mp4",
71         "uriAriaLabel": "Video location ? a gs:// bucket URL or a path on the engine host",
72         "hint": "Paste a gs:// bucket link, or a video path on the machine running the
engine.",
73         "submit": "Process video",
74         "submitting": "Submitting?",
75         "cancel": "Cancel",
76         "working": "Processing your video?",
77         "progress": "{progress}",
78         "retry": "Try again",
79         "failed": "Couldn't process the video: {error}",
80         "invalidUri": "Enter a video location first ? a gs:// link or a path on the
engine.",
81         "networkError": "Can't reach the engine. Is it running?",
82         "emptyResult": "Processing finished but found no rallies in this video. Try a
clearer match recording.",
83         "timeout": "This is taking longer than expected. The job may still be running ?
reload with the video link, or try again."
84     },
85     "selection": {
86         "count": "{count, plural, one {# rally selected} other {# rallies selected}}"
87     }
88 }
89

```

9. user (2026-06-21T14:34:13.137Z)

```

1     # Friendly & Multilingual ? non-tech-savvy UX + Hindi/Kannada localization
2
3     > **Status:** `DRAFT` ? owner-gated (implementation starts only on explicit approval) .
**Owner:** Avi Dullu . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
4     > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE).
5     > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; thread in `NEXT_STEPS.md`; pointer in engine `docs/DOC_STATUS.md`; memory
`current-state` + `next-session-ux-localization`.
6     > **Relation to existing docs:** this is the **implementation project** for two engine
design docs ? it *adapts & supersedes-in-practice* `badminton-highlight-
indexer/docs/I18N_PLAN.md` (localization) and the UX half of `~/docs/UI_PROPOSAL.md` (non-tech-
savvy console). Those stay the canonical *design*; this doc carries the *tracked execution*

```

against today's surfaces.

7 > **Honesty note:** each claim tagged `[verified]` (checked against code this session),
`[researched]` (needs sign-off), or `[design]` (proposed). Not legal advice ? the OFL font
question (D-FONT) is owner/counsel-gated.

8
9 ---
10
11 ## 0. TL;DR
12 A coach or parent in Bangalore should land on KhelSutra, feel at home, set up their
camera with confidence, and get highlights ? **in Kannada or Hindi**, with **zero hand-**
holding. This project isolates that goal into one tracked workstream-pair: **(A)** make the UI
super friendly for non-tech-savvy users, and **(B)** localize to Hindi + Kannada. Both surfaces
(`web/` React SPA + `site/` vanilla marketing) have **zero i18n today** `[verified]`, so the
foundation ? shared `locales/{en,kn,hi}` catalogs + a negotiation contract + CI parity ? is the
real deliverable; translations follow. **The single most important caveat:** translations are
human-reviewed (Gemini drafts are *inference only*, never training ? the standing
guardrail), and Kannada/Hindi rendering is **blocked** on the OFL font decision (D-FONT). The
cheapest unblocked high-impact first slice is the **SPA i18n scaffold** + language switcher (en-
only) ? it locks the shared catalog/negotiation contract and adds the render-test safety net
the SPA currently lacks.

13
14 ## 1. Problem & goal
15 **Why now.** The public site (kheelsutra.guru) is live, clean, honest, and tested. The
PMF beachhead is **Bangalore badminton academies**; many coaches, parents, and players are far
more comfortable in **Kannada** than English, and most are **not highly tech-savvy** `[verified`
? design intent]` (`UI_PROPOSAL.md:25,28`, `I18N_PLAN.md:10-13`, `PMF_MARKET_RESEARCH.md:116`,
`KHELSUTRA\_PRODUCT\_OVERVIEW.md:13/50/61`). An English-only, knob-heavy product adds friction at
the exact point of first contact.

16
17 **Concrete outcome** ("good" looks like): **a Kannada/Hindi-comfortable, phone-first, non-**
technical coach/parent can ? without help ? (1) understand what the product does and how to
record reliably, (2) request access / use the console, (3) pick rallies and get a reel, and (4)
share it on WhatsApp, **entirely in their language**.

18
19 **Read to get current:** this doc's §3?§4; the engine design `I18N\_PLAN.md` (canonical
i18n design) + `UI\_PROPOSAL.md` §2 (personas, the 6 alpha screens, "teach recording *before*
upload"); `VIDEO\_RECORDING\_GUIDELINES.md`; this repo's `PER\_RALLY\_SELECTION.md` (the honest SPA
scope) + `REQUEST\_ACCESS\_FORM.md`.

20
21 ## 2. Decisions locked (structural ? owner directive 2026-06-21)
22 | # | Decision | Source / date | Implication |
23 |---|-----|-----|-----|
24 | D1 | **Isolate as ONE tracked project** spanning both workstreams (A: UX, B: kn/hi
l10n) | owner, 2026-06-21 | This doc; §7 = source of truth; archived on DoD |
25 | D2 | **Home = `kheelsutra/docs/`** (the implementation repo) | owner-gated tracked-doc
convention | `web/` + `site/` both live here; §7 tracks kheelsutra PRs; engine P2 backend-l10n is
a cross-repo deferred row |
26 | D3 | **Adapt `I18N\_PLAN.md` to today's surfaces:** P1 (vanilla `t()` shim) ? `site/`;
P3 (react-i18next) ? `web/`; **drop the dead `backend/api/static` retrofit** | `[verified]` the
doc targets `backend/api/static` (`I18N\_PLAN.md:25,30,156,191`) which exists only in the engine
repo, not here | The doc's framework-agnostic *foundation* (catalog format + negotiation + ICU
plurals + CI) survives unchanged; only the wiring re-points |
27 | D4 | **Engine P2 backend-localization stays owner-gated + cross-repo** (not in this
DoD) | `CLAUDE.md` guardrails | Dynamic engine error/report text localizes later; tracked as a
deferred row |
28 | D5 | **Archive trigger = en+kn+hi live on BOTH surfaces** (+ the non-tech UX
baseline), then run the engine `DOC\_STATUS.md` §2 archival checklist | owner, 2026-06-21 | See
§9 |
29 | D6 | **D-LOCALES:** `en` + `kn` for v1, `hi` next; always-fallback to `en` | owner,
2026-06-21 | Kannada-first beachhead; `hi` = catalog-only (both 2-form CLDR) |
30 | D7 | **D-FONT:** OFL-for-media carve-out ? self-host Noto Sans Kannada/Devanagari
(counsel may still confirm) | owner, 2026-06-21 | Unblocks B4/B5/A3 + the kn/hi reel watermark;
an explicit exception to the MIT/Apache/BSD code guardrail (fonts = media) |
31 | D8 | **D-SURFACE-FIRST:** lead with the `web/` SPA scaffold; SKIP the dead static-UI
retrofit | owner, 2026-06-21 (via B1 go-ahead) | B1 is greenlit; site recording-education

```

localized early |
32 | D9 | **D-CATALOG-HOME: co-locate `locales/` + CI in `khelsutra`** (shared, consumed by
SPA + site) | owner, 2026-06-21 (via B1 go-ahead) | No cross-repo coupling |
33
34 *Only **D-BACKEND-DEPTH** (§8.3) remains open ? and it's a deferred, cross-repo engine
concern (B7), not a v1 blocker.*
35
36 ## 3. Foundation ? research / prior art
37 **Current-state facts `[verified]` this session** (adversarially checked against the
working tree):
38 - **web/` SPA: zero i18n.** `web/package.json` deps are only `react`/`react-dom`; no
`i18next`/`react-intl`/`formatjs`; no `locales/` dir; no `t()`/`useTranslation`/`data-i18n`
anywhere. User-facing English is hardcoded in `*.tsx` (e.g. `QueryBar.tsx:46-47,56,72,76,78`;
`App.tsx:115,211,224`) ? ~50 literals + ~16 derived labels.
39 - **site/` marketing: vanilla, zero i18n.** Three `**Recording-education is *not* greenfield.** The homepage "Record it right" gist
(`index.html:324-364`) + the `/capture` checklist (`capture.html`, routed `server.js:51`,
guarded `assets.test.js:94`) already ship the user-facing best-practices surface. *Unshipped:*
the empirical-threshold framing and the programmatic pre-flight validator (a code change, not a
page).
41 - **Persona is design intent, not built.** kn/hi + zero-knobs are **proposed/owner-
gated**; the app is English-only today (`KHELSUTRA_PRODUCT_OVERVIEW.md:61`). It's
Kannada-**first**, Hindi-**next** ? not Kannada-only.
42
43 **Localization prior art `[researched]`:
44 - **CLDR plural categories:** Hindi = `{one, other}`; Kannada = `{one, other}`. Both are
2-form (same shape as English) ? once ICU MessageFormat is in the catalog, adding `hi`/`kn`
plurals is **catalog-only**, no code. (ICU still required because English `count>1?'s':``` hand-
rolling ? present today in the SPA ? doesn't generalize.)
45 - **react-i18next** (web/): `i18next` + `react-i18next` + `i18next-browser-
languagedetector` + an ICU/plural post-processor; lazy-load per-locale JSON; persist via
`localStorage`.
46 - **Shared catalog + tiny shim** (site/): the SAME i18next-flavored JSON consumed by a
~30-line `t(key, vars)` shim that walks `data-i18n` attributes ? no build step, no framework,
mirrors i18next interpolation/plural rules for the 2-form languages.
47 - **Noto Sans Kannada / Noto Sans Devanagari** are **OFL-1.1**; OFL permits commercial
bundling/redistribution (fonts = media, not code/weights). Self-host (e.g. `@fontsource`) with
`font-display: swap` + subsetting. **Gated on D-FONT.**
48 - **Non-tech-savvy / HCI4D patterns:** icon **+ text** (never icon-only), non-color
selection indicators, large touch targets, mobile-first, WhatsApp as the share channel, a
prominent persistent language switcher.
49
50 ## 4. Design / architecture
51 **Two surfaces, one shared source of truth.**
52
53 ```
54 locales/                                # shared catalogs ? co-located in khelsutra (D-CATALOG-
HOME, see §8)
55   en/  common.json                      # source of truth ? devs author keys here
56   kn/  common.json                      # Kannada (Gemini-draft ? human-review)
57   hi/  common.json                      # Hindi (next ? the extensibility proof)
58
59 web/  ? react-i18next                    # I18N_PLAN P3: provider in main.tsx; useTranslation in
components
60 site/ ? t(key,vars) shim                 # I18N_PLAN P1 retargeted: data-i18n attrs over the
static HTML
61 ```
62
63 - **Negotiation contract (one order, both surfaces):** `persisted pref (localStorage) ?
switcher / ?lang=<code> ? navigator/Accept-Language ? "en"`. Supported locales in a single
module; anything else falls back to `en`; **a missing key degrades to English, never to a raw
key.**

```

64 - ****Keys, never prose**** ? `t("compile.button")`, with ICU for interpolation/plurals: ``{count, plural, one {# rally} other {# rallies}}``. This replaces the SPA's current hand-rolled ``Segment${count>1?'s':''}``.

65 - ****Translation pipeline (guardrail-clean):**** dev authors `en` keys ? ``scripts/i18n_draft.py``-style Gemini ****draft**** (inference only, never training; flags every value for review) ? ****owner/native-speaker review**** ? CI key-parity ? merge. Gemini never touches `en`, never invents keys.

66 - ****CI enforcement (keeps it "by design")**** (1) ****key-parity**** ? every locale has exactly the `en` key set (drop a stub `ta.json` ? CI lists every gap = the extensibility proof); (2) ****no-bare-strings**** ? flags user-facing literals not routed through `t()` (curated allowlist to start).

67 - ****Fonts**** bundle Noto Sans Kannada + Devanagari ****iff D-FONT = OFL carve-out****; this also unblocks a Kannada/Hindi ****reel watermark**** (engine ``WatermarkConfig.font_path`` already takes a path ? config, not code).

68

69 ****Non-tech-savvy UX levers**** (ranked impact × effort, grounded in personas; surface-tagged):

70

Lever	Surface	Impact	Effort	Note
Plain-language copy ? strip engine jargon, keep honest caveats	both	high	small	banners + footer carry backend vocabulary today
Localized **WhatsApp share** for the downloaded reel	both	high	small	WhatsApp is <i>the</i> channel for this audience
Language switcher persisted via a shared localStorage key	both	high	small	site promises kn/hi but ships no switcher
Localize **recording-education** (capture + gist) kn?hi	site	high	medium	gates input quality; the highest-leverage kn content
Mobile-first touch targets; tooltip-text ? visible	web	high	medium	tiny targets + desktop layout + invisible tooltips on phones
Icon **+ text** + non-color selection indicator	web	medium	small	never icon-only
Friendly engine-error mapping over raw FastAPI <code>`detail`</code>	web	medium	medium	<code>`CompileResult`</code> passes engine text untranslated
Plain-language framing for the Request-Access GPU/storage Qs	site	medium	small	"GPU"/"S3/GCS" are intimidating
Background-processing reassurance + status as icon chips	web	medium	small	honest expectations on slow uplinks
Fix English-source recording gaps (a diagram) before translating	site	medium	medium	the diagram is reusable across all 3 languages
Tap-only filter controls alongside the English query bar	web	medium	large	the <code>`parseQuery()`</code> parser is English-grammar only
Keep the honest **today-vs-roadmap** split through translation	both	medium	small	never soften "experimental"/"not available yet"

85

86 ****Reuse map (what's new vs reused):**** ****Reused**** ? the i18next-flavored JSON + negotiation contract + ICU plurals + key-parity/no-bare-strings CI + draft?review pipeline (all from ``I18N_PLAN.md``); the shipped recording pages; the SPA's ``lib/format.ts``/``lib/query.ts``; ``site/src/assets.test.js`` + the SPA's vitest harness. ****New**** ? ``locales/`` catalogs; the SPA react-i18next provider + switcher; the site ``t()`` shim + ``data-i18n`` wiring; Noto font bundle (gated); a jsdom render-test env for the SPA.

87

88 ****5. Privacy / scope note**

89 i18n touches ****presentation only**** ? no new personal-data collection, no biometrics, minors-excluded posture unchanged. WhatsApp share opens the user's own client with a pre-filled message; it sends nothing server-side. Gemini-draft translation is inference, not training (no user data in the loop).

90

91 ****6. Honest limits ? what this does NOT do**

92 - A green CI / DONE status proves ****key-parity + render + no-bare-strings****, ****not**** translation ****quality**** ? that's human-gated and owner-paced.

93 - Kannada/Hindi will ****not render**** until the Noto fonts land (D-FONT); until then kn/hi catalogs exist but display tofu/fallback.

94 - The SPA is ****job-tethered****; localizing ****dynamic engine errors/report text**** needs engine P2 (owner-gated, cross-repo, D-BACKEND-DEPTH) ? out of this DoD.

```

95     - The query bar stays **English-grammar parsing**; kn/hi users rely on tap-only controls
(a later, larger slice), not free-text Kannada queries.
96     - **Mobile-perfect** layout is an alpha non-goal (usable-on-phone is in scope; pixel-
perfect is not).
97
98     ## 6.5 Testing strategy ? automatic language negotiation (rigorous, owner-required)
99     **The bar (owner, #41):** a user whose **computer/browser language is Kannada or Hindi
must get the respective version automatically ** ? that is the core promise and it is **proven by
tests at every layer, never assumed.** Auto-selection IS the negotiation order (persisted pref ?
`?lang=` ? browser/`Accept-Language` ? `en`); each step gets a test.
100
101     **Negotiation matrix ? every row is an automated test:**
102
103     | # | Signal under test | Setup | Expected | Proven in |
104     |---|---|---|---|---|
105     | N1 | **Browser/OS = Kannada** | no persisted pref, no `?lang`,
`navigator.languages=["kn-IN", ?]` | resolves to **kn** ? Kannada content (once the kn catalog
ships) | **B1** detection . **B5** content |
106     | N2 | **Browser/OS = Hindi** | `navigator.languages=["hi-IN", ?]` | resolves to **hi**
? Hindi content | **B6** (hi added to the supported set + catalog) |
107     | N3 | **`?lang=` override** | `?lang=kn`, browser = en | kn | B1 |
108     | N4 | **Persisted pref wins** | `localStorage=kn`, browser = en, no `?lang` | kn (a
returning user keeps their choice) | B1 |
109     | N5 | **Region subtag normalizes** | `navigator.language="kn-IN"` | base **kn** (not a
"kn-IN" miss) | B1 |
110     | N6 | **Unsupported language** | `navigator="ta-IN"` (Tamil, not yet supported) |
**en** fallback ? never a locale we can't serve | B1 |
111     | N7 | **Missing key in active locale** | kn active, a key absent from `kn/common.json`
| **en** fallback for that key (no raw key shown) | B1/B5 |
112     | N8 | **Static site (vanilla shim)** | `Accept-Language: kn` / `?lang=kn` on a `site/`
page | kn-rendered page (same order as the SPA) | B3 |
113     | N9 | **Backend (engine P2)** | `Accept-Language: kn` to the API | kn customer/report
text | B7 (owner-gated) |
114
115     **Levels & tooling:**
116     - **SPA (`web/`):** jsdom tests stub `navigator.language`/`navigator.languages`,
`localStorage`, and the `?lang` query ? assert `i18n.language`/`resolvedLanguage` + rendered
text per the matrix. *(B1 lands N1/N3?N7 at the **detection** level now ? `kn` is supported; the
kn/hi **content** assertions ride the catalogs in B5/B6.)*
117     - **Static site (B3):** the shared `t()` shim's negotiation is unit-tested the same way;
plus a server test that `Accept-Language: kn`/`?lang=kn` returns the kn-rendered page.
118     - **CI key-parity:** guarantees every locale has every key, so a correctly-negotiated
locale can never fall through to a raw key.
119     - **End-to-end (manual, owner machine ? per I18N_PLAN $6):** set the real OS/browser to
Kannada, then Hindi ? load both surfaces ? confirm the right language renders **and the Noto
fonts display** (the cloud CI container can't render fonts; the owner eyeball is the final
gate).
120     - **? Query parsing / semantic understanding (forward-looking, owner directive
2026-06-21) ? i18n testing is MANDATORY there.** The query bar is **English-grammar-only** today
(a documented limit, tracker A9): the user types English queries (`over 10s`, `top 15 longest`)
and a localized **sample-query prompt** shows them what to type in **every** locale (kn/hi
included). The **moment** the parser gains **semantic understanding** or accepts **Kannada/Hindi
query input** (e.g. typing a Kannada/Hindi phrase, or a conversational/NL builder ? I18N_PLAN
P4), i18n tests become a **merge gate**: a kn/hi query (typed or spoken) must parse to the
**correct selection**, the localized **parse-feedback** ("parsed-pill") must render in the
active locale, and the unsupported/empty-result messages must localize. **No query-NL /
semantic-parsing work merges without those tests.**
121
122     A locale that's reachable by the switcher but **not auto-selected from the browser
language is a bug, not a "later".** This matrix is a DoD gate ($9).
123
124     ## 7. Deliverables & progress tracker ? **source of truth**
125     Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking.
126

```

127 | ID | Deliverable | Workstream | Depends on | Gated? | Status | PR |
128 |----|-----|-----|-----|-----|-----|----|
129 | P0 | This tracked project doc + inbound pointers | ? | ? | No | ? |
[41](https://github.com/avidullu/khelsutra/pull/41) . engine
[275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275) |
130 | B1 | **SPA i18n scaffold + language switcher, en-only** (react-i18next + detector +
`en` catalog + switcher + persistence + **jsdom render tests** + **browser-language negotiation
tests** §6.5 N1/N3?N7) | B | P0 | No | ? | [42](https://github.com/avidullu/khelsutra/pull/42)
|
131 | B2 | Extract SPA strings ? `en` catalog (~50 literals + ~16 derived; ICU plural keys);
add **key-parity + no-bare-strings CI** | B | B1 | No | ? |
[45](https://github.com/avidullu/khelsutra/pull/45) |
132 | A1 | Plain-language copy pass on the `en` catalog (de-jargon; icon+text; non-color
selection) | A | B2 | No | ? | ? |
133 | B3 | Tiny shared `t()` shim + `data-i18n` plumbing for `site/`, en-only (proves shared
catalogs across surfaces) | B | B2 | D-CATALOG-HOME | ? | ? |
134 | A2 | Localized **WhatsApp share** for the reel (SPA + site) | A | B1/B3 | No | ? | ? |
135 | A6 | Plain-language framing for Request-Access GPU/storage questions | A | B3 | No | ?
| ? |
136 | A7 | Fix English-source recording gaps (add a placement diagram) before localizing | A
| ? | No | ? | ? |
137 | B4 | **Noto Sans Kannada + Devanagari** bundle (OFL carve-out) ? *SPA done; reel
watermark `font\_path` = engine follow-up* | B | ? | D-FONT ? | ? |
[46](https://github.com/avidullu/khelsutra/pull/46) |
138 | B5 | **Kannada (`kn`) catalog** ? render ? *SPA done (AI-draft, native review
pending); `site/` rides B3* | B | B2/B4 | D-LOCALES ? | ? |
[47](https://github.com/avidullu/khelsutra/pull/47) |
139 | A3 | Localize **recording-education** (capture + homepage gist) to kn | A | B3/B5 |
D-LOCALES+D-FONT | ? | ? |
140 | A4 | Mobile-first touch targets + tooltip-text ? visible (SPA) | A | B1 | No | ? | ? |
141 | A5 | Friendly engine-error mapping over raw FastAPI `detail` (SPA, client-side map) |
A | B2 | No | ? | ? |
142 | B6 | **Hindi (`hi`) catalog** ? render (the extensibility proof ? just a catalog) ?
SPA done (AI-draft, native review pending) | B | B5 | D-LOCALES ? | ? |
[48](https://github.com/avidullu/khelsutra/pull/48) |
143 | A8 | Background-processing reassurance + verdict/status as icon chips (SPA) | A | B2 |
No | ? | ? |
144 | A9 | Tap-only filter controls alongside the English query bar (SPA) | A | B2 | No | ?
| ? |
145 | B7 | *(deferred, cross-repo)* Engine **P2 backend customer-facing localization**
(report `customer\_` + rejection reasons) | B | engine | **D-BACKEND-DEPTH, owner-gated** | ? |
? |
146
147 *** MILESTONE ? SPA fully trilingual (2026-06-21):** the `web/` Studio SPA now speaks
en + kn + hi end to end (B1?B2?B4?B5?B6, all merged). A **Kannada-default browser auto-
renders Kannada**, and the switcher walks kn ? hi ? en ? **proven by a Playwright screencast**
with `locale: kn-IN` (`khelsutra-i18n-screencast.mp4`; frames captured per locale + `
lang>` tracked). This meets the owner's success criterion **on the SPA surface**. **Still open
for the full §9 DoD:** **B3** (the `site/` marketing surface via the shared `t()` shim ? the
homepage/capture pages are still English-only), **A3** (recording-education in kn/hi), the
non-tech UX A-items, and a **native-speaker review** of the AI-drafted kn/hi catalogs before
a wide public launch. Do **not** archive until both surfaces are trilingual.
148
149 ## 8. Open questions ? owner (blocking gates) `[design]`
150 *Status (2026-06-21): 4 of 5 **LOCKED/ADOPTED** by the owner ? see §2 (D6?D9). Only
D-BACKEND-DEPTH remains open (and it's deferred, not a v1 blocker).*
151 1. **D-FONT ? ? LOCKED ? (a) OFL-for-media carve-out, self-host Noto Sans
Kannada/Devanagari** (counsel may still confirm). An explicit exception to the MIT/Apache/BSD
code guardrail (fonts = media). Unblocks B4/B5/A3 + the kn/hi reel watermark.
152 2. **D-LOCALES ? ? LOCKED ? `en` + `kn` for v1, `hi` next, always-fallback to `en`**
(both 2-form CLDR ? `hi` is catalog-only).
153 3. **D-BACKEND-DEPTH ? ? OPEN (deferred, not a v1 blocker).** Recommend **customer
verdict/report `customer\_` + validation-rejection reasons first; technical/operator text stays
English; phase two** (engine repo, owner-gated ? tracker row B7).
154 4. **D-SURFACE-FIRST ? ? ADOPTED ? lead with the SPA scaffold (B1), localize site

recording-education early, SKIP the dead static-UI retrofit.**

155 5. **D-CATALOG-HOME ? ? ADOPTED ? co-locate `locales/` + CI in `khelsutra`** (a shared i18n dir consumed by both the SPA and the site). (The engine `I18N\_PLAN` assumed the indexer repo; this is the adaptation.)

156

157 ## 9. Definition of done

158 - [] **Automatic language negotiation (§6.5) is green** ? a **Kannada browser gets Kannada and a Hindi browser gets Hindi** (N1/N2); persisted-pref / `?lang` / region-subtag / unsupported-fallback / missing-key-fallback all proven (N3?N7) on the SPA, and the static-site shim honors `Accept-Language` + `?lang` (N8).

159 - [] **Manual cross-check (owner machine):** OS/browser set to Kannada and to Hindi each render the respective language on BOTH surfaces, with Noto fonts displaying.

160 - [] `en + kn + hi` catalogs live and **rendering** on BOTH `web/` SPA and `site/` marketing (switcher + persistence + `en` fallback; no raw keys shown).

161 - [] **Noto Sans Kannada + Devanagari render correctly** ? owner eyeball on a real machine.

162 - [] CI **key-parity + no-bare-strings** green; a stub locale flags every gap.

163 - [] **Recording-education** (capture + homepage gist) available in `kn` + `hi`.

164 - [] **Non-tech UX baseline:** plain-language copy (no raw engine jargon in user-facing strings), mobile-usable touch targets, icon+text, WhatsApp share, honest today-vs-roadmap preserved through translation.

165 - [] §7 all rows ? (or explicitly deferred with a reason) ? flip **Status ? DONE** + completion note ? run engine `DOC\_STATUS.md §2` archival checklist ? `git mv` to `docs/archives/`.

166

167 ## 10. References

168 **Internal code anchors `[verified]`:** `web/package.json`, `web/src/{App.tsx,main.tsx,components/QueryBar.tsx,components/CompileResult.tsx,lib/query.ts}`; `site/public/{index,capture,own-model}.html`, `site/server.js`, `site/src/assets.test.js`.

This repo's docs: `docs/PER\_RALLY\_SELECTION.md`, `docs/REQUEST\_ACCESS\_FORM.md`, `docs/README.md`, `NEXT\_STEPS.md`. **Engine design docs (canonical):** `badminton-highlight-indexer/docs/{I18N\_PLAN.md,UI\_PROPOSAL.md,VIDEO\_RECORDING\_GUIDELINES.md,PMF\_MARKET\_RESEARCH.md,PMF\_FIELDED\_SIGNALS.md,KHELSUTRA\_PRODUCT\_OVERVIEW.md,DOC\_STATUS.md}`. **External `[researched]`:** Unicode CLDR plural rules (kn/hi = one/other); react-i18next + i18next-icu; SIL Open Font License 1.1; Google Noto Sans Kannada/Devanagari.

169

170 ---

171

172 ### Changelog

173 - `2026-06-21` ? created. Isolated the UX + Hindi/Kannada work into one tracked project (owner directive). Grounded in a multi-agent research+verification pass (web/+site/ string maps, Indic-l10n prior art, 5 adversarial verdicts). Decisions D1?D5 locked; 5 product/licensing decisions surfaced as open; §7 tracker seeded with B1 as the recommended first slice.

174 - `2026-06-21` ? owner locked **D6 (en+kn?hi)** + **D7 (OFL Noto carve-out)** and adopted **D8 (SPA-first)** + **D9 (catalogs in khelsutra)**; only D-BACKEND-DEPTH remains open. **B1 greenlit ? ? in progress.**

175 - `2026-06-21` ? **B1 opened ? [#42](https://github.com/avidullu/khelsutra/pull/42)**: SPA i18n scaffold (react-i18next + ICU + detector) + `en` catalog + language switcher + the SPA's first jsdom render tests. Verified typecheck/79-tests/build green + live in-browser (switcher persists, en-fallback). Doc/pointer PRs opened: khelsutra [#41](https://github.com/avidullu/khelsutra/pull/41), engine [#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275).

176 - `2026-06-21` ? **§6.5 Testing strategy added (owner ask, #41):** a rigorous **automatic-negotiation matrix** (N1?N9) so a Kannada/Hindi **browser** auto-gets the respective version ? proven at the detection level now (B1), at the content level with the catalogs (B5/B6), on the static shim (B3), and by a manual font-render eyeball. Wired into the DoD (§9). B1 (#42) gains the detection tests (N1/N3?N7).

177 - `2026-06-21` ? **? SPA MILESTONE: fully trilingual (owner directive "take it to completion + screencast").** B2 (#45, full string extraction + key-parity/no-bare-strings CI) ? B4 (#46, self-host OFL Noto) ? B5 (#47, Kannada) ? B6 (#48, Hindi ? the extensibility proof) all merged. 99 web tests green incl. N1/N2 browser-detection + B5/B6 render + key-parity. **Playwright screencast** (`locale: kn-IN`): a Kannada-default browser auto-renders Kannada ? switch hi ? switch en, each with `` tracked (`khelsutra-i18n-screencast.mp4` + per-locale frames). kn/hi are **AI-drafted ? native review pending**. **Remaining for full §9 DoD:** B3 (the `site/` marketing surface), A3 + the UX A-items, native-speaker translation review.

10. assistant (2026-06-21T14:34:17.531Z)

Now let me read the three site HTML files and the engine I18N_PLAN.md P1 section. Let me also check the existing site test files and locale catalogs structure.

11. user (2026-06-21T14:34:18.452Z)

```

1      <!doctype html>
2      <html lang="en">
3      <head>
4      <meta charset="utf-8" />
5      <meta name="viewport" content="width=device-width, initial-scale=1" />
6      <title>KhelSutra ? every rally indexed, the dead time gone</title>
7      <meta name="description" content="KhelSutra turns a full badminton match into a clean
index of rallies with the dead time stripped out ? a highlight reel you can watch and download.
Invite-only alpha, Bengaluru." />
8      <meta property="og:title" content="KhelSutra ? every rally indexed, the dead time gone"
/>
9      <meta property="og:description" content="A full match becomes a clean highlight reel ?
only the real play, no dead time. Invite-only alpha for academies and players in Bengaluru." />
10     <meta property="og:type" content="website" />
11     <meta property="og:url" content="https://khelsutra.guru" />
12     <meta property="og:image" content="https://khelsutra.guru/og.png" />
13     <meta property="og:image:width" content="1200" />
14     <meta property="og:image:height" content="630" />
15     <meta name="twitter:card" content="summary_large_image" />
16     <meta name="twitter:title" content="KhelSutra ? every rally indexed, the dead time gone"
/>
17     <meta name="twitter:description" content="A full match becomes a clean highlight reel ?
only the real play, no dead time. Invite-only alpha, Bengaluru." />
18     <meta name="twitter:image" content="https://khelsutra.guru/og.png" />
19     <meta name="theme-color" content="#0C261F" />
20     <link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg'
viewBox='0 0 32 32'%3E%3Crect width='32' height='32' rx='8' fill='%230C261F'%3E%3Cpath d='M13
10 L23 16 L13 22 Z' fill='%2319B36B'%3E%3C/svg%3E" />
21     <link rel="preconnect" href="https://fonts.googleapis.com" />
22     <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
23     <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;
700;800&display=swap" rel="stylesheet" />
24     <style>
25     :root{
26         --ink:#0C261F; --ink2:#123A2E; --green:#19B36B; --green-d:#149A5C;
27         --lime:#C7F2B0; --paper:#F6F8F4; --card:#fff;
28         --tx:#14271F; --mut:#566B61; --line:#E4E9E2;
29         --on-dark:#EAF3EE; --on-dark-mut:#A7C3B7;
30         --r:14px; --rs:10px; --maxw:1080px;
31         --font:'Plus Jakarta Sans',system-ui,-apple-system, Segoe
UI,Roboto,Helvetica,Arial,sans-serif;
32     }
33     *{box-sizing:border-box}
34     html{scroll-behavior:smooth}
35     body{margin:0;font-family:var(--font);color:var(--tx);background:var(--paper);line-
height:1.65;-webkit-font-smoothing:antialiased}
36     a{color:inherit}
37     .wrap{max-width:var(--maxw);margin:0 auto;padding:0 22px}
38     h1,h2,h3{line-height:1.15;margin:0;letter-spacing:-.02em}
39     h1{font-size:clamp(2rem,5vw,3.4rem);font-weight:800}
40     h2{font-size:clamp(1.5rem,3.2vw,2.2rem);font-weight:700}
41     h3{font-size:1.15rem;font-weight:700}
42     p{margin:.6rem 0}
43     .btn{display:inline-flex;align-items:center;gap:.5rem;font-weight:600;font-
size:1rem;border-radius:999px;padding:.78rem 1.4rem;text-decoration:none;border:1px solid
transparent;cursor:pointer;transition:.15s}

```

```

44     .btn-primary{background:var(--green);color:#04140D}
45     .btn-primary:hover{background:var(--green-d)}
46     .btn-ghost{background:transparent;border-color:rgba(255,255,255,.28);color:var(--on-
dark)}
47     .btn-ghost:hover{border-color:rgba(255,255,255,.6)}
48     .btn-dark{background:var(--ink);color:#fff}
49     .btn-dark:hover{background:var(--ink2)}
50     .btn[disabled]{opacity:.75;cursor:progress}
51     /* in-flight spinner for the request-access submit */
52     .spin{display:inline-block;width:1em;height:1em;border:2px solid currentColor;border-
right-color:transparent;border-radius:50%;animation:spin .6s linear infinite}
53     @keyframes spin{to{transform:rotate(360deg)}}
54     .eyebrow{display:inline-flex;align-items:center;gap:.5rem;font-size:.82rem;font-
weight:600;letter-spacing:.02em;
55         background:rgba(25,179,107,.14);color:#0a6c45;border:1px solid
rgba(25,179,107,.3);padding:.3rem .8rem;border-radius:999px}
56
57     /* header */
58     header{position:sticky;top:0;z-index:20;background:rgba(246,248,244,.86);backdrop-
filter:blur(10px);border-bottom:1px solid var(--line)}
59     .nav{display:flex;align-items:center;justify-content:space-between;height:66px}
60     .brand{display:flex;align-items:center;gap:.6rem;font-weight:800;font-
size:1.18rem;letter-spacing:-.02em;text-decoration:none}
61     .brand .dot{color:var(--green)}
62     .nav-links{display:flex;align-items:center;gap:1.6rem}
63     .nav-links a{font-size:.95rem;font-weight:500;color:var(--mut);text-decoration:none}
64     .nav-links a:hover{color:var(--tx)}
65     .brand .play{display:inline-block;width:0;height:0;border-style:solid;border-width:7px
0 7px 12px;border-color:transparent transparent transparent var(--green);margin:0
.32em;transform:translateY(1px)}
66
67     /* hero */
68     .hero{background:linear-gradient(160deg,var(--ink) 0%,var(--ink2)
100%);color:var(--on-dark);position:relative;overflow:hidden}
69     .hero .wrap{padding-top:74px;padding-bottom:88px;position:relative;z-index:2}
70     .hero .eyebrow{background:rgba(199,242,176,.12);color:var(--lime);border-
color:rgba(199,242,176,.25)}
71     .hero h1{margin:1.1rem 0 0;max-width:15ch}
72     .hero h1 .hl{color:var(--lime)}
73     .hero .lead{font-size:clamp(1.05rem,1.6vw,1.28rem);color:var(--on-dark-mut);max-
width:54ch;margin:1.2rem 0 0}
74     .hero-cta{display:flex;flex-wrap:wrap;gap:.8rem;margin-top:2rem}
75     .hero-note{margin-top:1.1rem;font-size:.86rem;color:var(--on-dark-mut)}
76     .shuttle{position:absolute;right:-60px;top:-40px;width:520px;max-
width:55%;opacity:.16;z-index:1;pointer-events:none}
77     .loop{display:flex;flex-wrap:wrap;gap:.6rem;margin-top:2.4rem}
78     .loop .chip{display:flex;align-
items:center;gap:.5rem;background:rgba(255,255,255,.06);border:1px solid rgba(255,255,255,.12);
79         border-radius:999px;padding:.5rem 1rem;font-weight:500;font-
size:.92rem;color:var(--on-dark)}
80     .loop .chip b{color:var(--lime);font-weight:700}
81     .loop .arrow{color:var(--on-dark-mut);align-self:center}
82
83     .hero-grid{display:grid;grid-template-columns:1.04fr .96fr;gap:46px;align-
items:center}
84     .hero-left{min-width:0}
85     .hero-visual{min-width:0}
86     .reel-card{background:#fff;border-radius:18px;border:1px solid
rgba(255,255,255,.14);box-shadow:0 26px 60px -24px rgba(0,0,0,.55);overflow:hidden}
87     .reel-head{display:flex;align-items:center;gap:.55rem;padding:.7rem .9rem;border-
bottom:1px solid var(--line)}
88     .reel-head .dots{display:flex;gap:5px}
89     .reel-head .dots i{width:9px;height:9px;border-
radius:50%;background:#D9DED7;display:block}
90     .reel-head .file{font-size:.78rem;color:var(--mut);font-weight:600}

```

```

91      .reel-video{position:relative;aspect-ratio:16/9;background:linear-
gradient(150deg,#0E3A2C,#17533C);display:flex;align-items:center;justify-content:center}
92      .reel-video .court{position:absolute;inset:16px;border:2px solid
rgba(199,242,176,.30);border-radius:6px}
93      .reel-video .court:before{content:"";position:absolute;left:50%;top:0;bottom:0;width:2
px;background:rgba(199,242,176,.30)}
94      .reel-video .court:after{content:"";position:absolute;top:50%;left:0;right:0;height:2p
x;background:rgba(199,242,176,.22)}
95      .reel-video .play{width:60px;height:60px;border-
radius:50%;background:var(--lime);display:flex;align-items:center;justify-
content:center;color:#0C261F;font-size:1.4rem;z-index:2;padding-left:4px;box-shadow:0 10px 26px
rgba(0,0,0,.35)}
96      .reel-video .badge{position:absolute;top:12px;left:12px;background:rgba(12,38,31,.72);
color:var(--lime);font-size:.7rem;font-weight:700;letter-spacing:.06em;padding:.26rem
.6rem;border-radius:999px;z-index:2}
97      .reel-video
.wm{position:absolute;bottom:10px;right:12px;color:rgba(234,243,238,.78);font-size:.66rem;font-
weight:500;z-index:2}
98      .reel-body{padding:.95rem 1.05rem 1.1rem}
99      .reel-body .lbl{font-size:.72rem;font-weight:600;letter-spacing:.04em;text-
transform:uppercase;color:var(--mut)}
100     .reel-track{position:relative;height:12px;border-
radius:6px;background:#E8ECE6;margin:.55rem 0 .95rem}
101     .reel-track span{position:absolute;top:0;bottom:0;background:var(--green);border-
radius:6px}
102     .reel-stats{display:flex;gap:10px}
103     .reel-stats .m{flex:1;background:var(--paper);border-radius:10px;padding:.6rem .85rem}
104     .reel-stats .m b{display:block;font-size:1.25rem;color:var(--tx);font-weight:700}
105     .reel-stats .m em{font-size:.76rem;color:var(--mut);font-style:normal}
106     .ask{display:flex;align-items:center;gap:.55rem;margin:.85rem 1.05rem
.2rem;padding:.6rem .7rem;border:1px solid var(--line);border-
radius:12px;background:var(--paper)}
107     .ask-ic{color:var(--green-d);font-size:.95rem}
108     .ask-tx{flex:1;font-size:.88rem;color:var(--tx);font-weight:500;line-height:1.3}
109     .ask-go{width:26px;height:26px;border-
radius:8px;background:var(--green);color:#04140D;display:flex;align-items:center;justify-
content:center;font-size:.85rem;flex:none}
110     .lbl{display:flex;justify-content:space-between;align-items:center}
111     .lbl .match{color:var(--green-d);font-weight:700}
112     .reel-track span.off{background:#D7DDD4}
113     .reel-track span.on{background:var(--green)}
114     .result{display:flex;align-items:center;gap:.7rem;margin-top:.2rem;padding:.65rem
.8rem;background:var(--ink);border-radius:12px}
115     .result .play{width:34px;height:34px;border-
radius:50%;background:var(--lime);color:#0C261F;display:flex;align-items:center;justify-
content:center;flex:none;font-size:.95rem;padding-left:2px}
116     .result-tx{flex:1}
117     .result-tx b{display:block;font-size:.84rem;font-weight:700;color:#fff}
118     .result-tx em{display:block;font-style:normal;font-size:.73rem;color:var(--on-dark-
mut)}}
119     .result .dl{color:var(--on-dark-mut);font-size:1.1rem}
120     .askc .q{font-weight:600;background:var(--paper);border:1px solid var(--line);border-
radius:10px;padding:.6rem .75rem;font-size:.95rem}
121     .askc .q i{color:var(--green-d);font-style:normal;margin-right:.35rem}
122     .askc .a{display:flex;gap:.5rem;color:var(--mut);font-size:.93rem;margin-
top:.75rem;padding-left:.1rem}
123     .askc .a i{color:var(--green-d);font-style:normal;margin-top:.1rem}
124     .ask-note{margin-top:1.5rem;font-size:.95rem;color:var(--mut)}
125
126     /* sections */
127     section{padding:72px 0}
128     .sec-head{max-width:60ch}
129     .sec-head .kicker{font-size:.82rem;font-weight:700;letter-spacing:.08em;text-
transform:uppercase;color:var(--green-d)}
130     .sec-head h2{margin:.5rem 0 0}

```

```

131     .sec-head p{color:var(--mut);font-size:1.08rem;margin-top:.7rem}
132
133     .grid{display:grid;gap:18px;margin-top:2.4rem}
134     .g4{grid-template-columns:repeat(4,1fr)}
135     .g3{grid-template-columns:repeat(3,1fr)}
136     .g2{grid-template-columns:repeat(2,1fr)}
137     .card{background:var(--card);border:1px solid var(--line);border-
radius:var(--r);padding:1.4rem}
138     .card .ic{width:42px;height:42px;border-radius:11px;display:flex;align-
items:center;justify-content:center;
139         background:rgba(25,179,107,.12);color:var(--green-d);font-size:1.3rem;margin-
bottom:.9rem}
140     .card h3{font-size:1.05rem}
141     .card p{color:var(--mut);font-size:.96rem;margin:.4rem 0 0}
142
143     .split{display:grid;grid-template-columns:1.1fr .9fr;gap:42px;align-items:center}
144     .panel{background:var(--ink);color:var(--on-dark);border-radius:18px;padding:2.2rem}
145     .panel h2{color:#fff}
146     .panel p{color:var(--on-dark-mut)}
147     .panel .pill{display:inline-
block;background:rgba(199,242,176,.14);color:var(--lime);font-size:.8rem;font-weight:600;
148         padding:.3rem .7rem;border-radius:999px;margin-bottom:1rem}
149
150     .road{display:grid;grid-template-columns:repeat(5,1fr);gap:14px;margin-top:2.2rem}
151     .road .step{background:var(--card);border:1px solid var(--line);border-
radius:var(--rs);padding:1.1rem}
152     .road .step .n{font-size:1.4rem;color:var(--green-d)}
153     .road .step h3{font-size:1rem;margin-top:.5rem}
154     .road .step p{font-size:.88rem;color:var(--mut);margin-top:.3rem}
155
156     .honest{background:#fff;border:1px solid var(--line);border-left:4px solid
var(--green);border-radius:0 var(--rs) var(--rs) 0;padding:1.2rem 1.4rem;margin-top:1.6rem}
157     .honest b{color:var(--tx)}
158     .honest p{color:var(--mut);margin:.3rem 0 0}
159
160     .notes{list-style:none;padding:0;margin:2rem 0 0;display:grid;grid-template-
columns:repeat(2,1fr);gap:12px}
161     .notes li{display:flex;gap:.6rem;background:var(--card);border:1px solid
var(--line);border-radius:var(--rs);padding:.9rem 1.1rem;color:var(--mut);font-size:.95rem}
162     .notes li i{color:var(--green-d);margin-top:.15rem}
163
164     details{background:var(--card);border:1px solid var(--line);border-
radius:var(--rs);padding:0 1.2rem;margin-top:10px}
165     details summary{cursor:pointer;font-weight:600;padding:1rem 0;list-
style:none;display:flex;justify-content:space-between;align-items:center;gap:1rem}
166     details summary::-webkit-details-marker{display:none}
167     details summary::after{content:"+";color:var(--green-d);font-size:1.3rem;font-
weight:400}
168     details[open] summary::after{content:"?"}
169     details p{color:var(--mut);margin:0 0 1.1rem}
170
171     .cta{background:linear-gradient(160deg,var(--ink) 0%,var(--ink2) 100%);color:var(--on-
dark);border-radius:20px;padding:3rem 2.4rem;text-align:center;margin:0 0 0}
172     .cta h2{color:#fff;max-width:20ch;margin:0 auto}
173     .cta p{color:var(--on-dark-mut);max-width:46ch;margin:.8rem auto 0}
174     .cta .hero-cta{justify-content:center}
175     .af{max-width:560px;margin:2.2rem auto 0;background:var(--card);border:1px solid
var(--line);border-radius:var(--r);padding:1.6rem 1.5rem;text-align:left}
176     .af .row{margin-bottom:1.05rem}
177     .af label{display:block;font-size:.9rem;font-weight:600;color:var(--tx);margin-
bottom:.4rem}
178     .af .hint{font-weight:400;color:var(--mut);font-size:.84rem}
179     .af input[type=text],.af input[type=email],.af select,.af
textarea{width:100%;font:inherit;font-
size:.95rem;color:var(--tx);background:var(--paper);border:1px solid var(--line);border-

```

```

radius:10px;padding:.6rem .75rem}
180     .af textarea{min-height:84px;resize:vertical}
181     .af input:focus,.af select:focus,.af textarea:focus{outline:none;border-
color:var(--green);box-shadow:0 0 3px rgba(25,179,107,.15)}
182     .seg{display:flex;gap:10px}
183     .seg label{flex:1;margin:0;border:1px solid var(--line);border-
radius:10px;padding:.7rem .8rem;cursor:pointer;font-weight:500;text-align:center}
184     .seg input{position:absolute;opacity:0}
185     .seg label:hover{border-color:var(--green)}
186     .seg label:has(input:checked){border-
color:var(--green);background:rgba(25,179,107,.07);color:var(--green-d);font-weight:700}
187     .opts{display:flex;flex-wrap:wrap;gap:8px}
188     .opts label{margin:0;display:inline-flex;align-items:center;gap:.4rem;border:1px solid
var(--line);border-radius:999px;padding:.4rem .8rem;font-weight:500;font-
size:.88rem;cursor:pointer}
189     .opts label:has(:checked){border-
color:var(--green);background:rgba(25,179,107,.07);color:var(--green-d)}
190     .opts input{accent-color:var(--green)}
191     .af .consent{display:flex;align-items:flex-start;gap:.5rem}
192     .af .consent input{margin-top:.2rem;accent-color:var(--green)}
193     .af .acad{display:none}
194     .af.is-academy .acad{display:block}
195     .hp{position:absolute;left:-9999px;width:1px;height:1px;overflow:hidden}
196     .form-done{max-width:560px;margin:2rem auto 0;text-
align:center;background:rgba(25,179,107,.08);border:1px solid
rgba(25,179,107,.3);color:var(--green-d);border-radius:var(--r);padding:1.4rem;font-weight:600}
197
198     footer{border-top:1px solid var(--line);padding:42px 0;color:var(--mut);font-
size:.9rem}
199     .foot{display:flex;flex-wrap:wrap;justify-content:space-between;gap:1.2rem;align-
items:center}
200     .foot a{color:var(--green-d);text-decoration:none}
201     .foot a:hover{text-decoration:underline}
202     .stamp{font-size:.82rem;color:var(--mut)}
203
204     @media (max-width:860px){
205         .nav-links{display:none}
206         .g4{grid-template-columns:repeat(2,1fr)}
207         .g3,.g2{grid-template-columns:1fr}
208         .split{grid-template-columns:1fr;gap:24px}
209         .road{grid-template-columns:repeat(2,1fr)}
210         .notes{grid-template-columns:1fr}
211         .shuttle{opacity:.1}
212         .hero-grid{grid-template-columns:1fr;gap:30px}
213         .reel-card{max-width:460px}
214     }
215
216     /* tool demo + sample reel ? revealed AFTER sign-up, then auto-closes */
217     .reelwrap{position:relative;max-width:680px;margin:.5rem auto 0;aspect-
ratio:16/9;border-radius:14px;overflow:hidden;border:1px solid var(--line);background:#0C261F}
218     .reelwrap iframe{width:100%;height:100%;border:0;display:block}
219     .tool-vid{display:block;width:100%;max-width:680px;margin:.5rem auto 0;border-
radius:14px;border:1px solid var(--line);background:#0C261F}
220     .reveal-h{max-width:680px;margin:1.2rem auto .15rem;font-size:.74rem;font-
weight:700;letter-spacing:.06em;text-transform:uppercase;color:var(--green-d);text-align:left}
221     .reveal-bar{max-width:680px;margin:.2rem auto 0;display:flex;justify-content:flex-end}
222     .reveal-x{background:transparent;border:1px solid var(--line);color:var(--mut);font-
size:.78rem;font-weight:600;padding:.32rem .72rem;border-radius:999px;cursor:pointer}
223     .reveal-x:hover{color:var(--tx);border-color:var(--mut)}
224 </style>
225 </head>
226 <body>
227
228 <header>
229     <div class="wrap nav">

```

```

230     <a class="brand" href="#top">Khel<span class="play" aria-
hidden="true"></span>Sutra</a>
231     <nav class="nav-links" aria-label="Primary">
232         <a href="#how">How it works</a>
233         <a href="#diff">Why it's hard</a>
234         <a href="/own-model">Own your model</a>
235         <a href="/flyer">Flyer</a>
236         <a href="#roadmap">Roadmap</a>
237         <a href="#faq">FAQ</a>
238     </nav>
239     <a class="btn btn-dark" href="#join">Request access</a>
240 </div>
241 </header>
242
243 <a id="top"></a>
244 <section class="hero" style="padding:0">
245     <div class="wrap hero-grid">
246         <div class="hero-left">
247             <span class="eyebrow">Invite-only alpha · Bengaluru</span>
248             <h1>Every rally, indexed. <span class="hl">The dead time, gone.</span></h1>
249             <p class="lead">KhelSutra finds every rally in your session and strips out the
knock-ups, walks and gaps ? so a full match becomes a clean highlight reel you can watch and
download. No editing, no scrubbing.</p>
250             <div class="hero-cta">
251                 <a class="btn btn-primary" href="#join">Request alpha access</a>
252                 <a class="btn btn-ghost" href="#how">See how it works</a>
253                 <a class="btn btn-ghost" href="#join">See it in action + a real reel</a>
254             </div>
255             <p class="hero-note">A phone or GoPro on a tripod is all you need. No special
hardware.</p>
256             <div class="loop">
257                 <span class="chip"><b>1.</b> Auto-index rallies</span>
258                 <span class="arrow">?</span>
259                 <span class="chip"><b>2.</b> Dead time removed</span>
260                 <span class="arrow">?</span>
261                 <span class="chip"><b>3.</b> Download your reel</span>
262             </div>
263         </div>
264         <div class="hero-visual">
265             <div class="reel-card">
266                 <div class="reel-head">
267                     <span class="dots"><i></i><i></i><i></i><i></i></span>
268                     <span class="file">match_2026.mp4 · indexed</span>
269                 </div>
270                 <div class="reel-body">
271                     <div class="lbl"><span>Rallies detected</span><span class="match">44
found</span></div>
272                     <div class="reel-track">
273                         <span class="on" style="left:2%;width:8%"></span>
274                         <span class="on" style="left:13%;width:6%"></span>
275                         <span class="on" style="left:22%;width:12%"></span>
276                         <span class="on" style="left:37%;width:7%"></span>
277                         <span class="on" style="left:47%;width:11%"></span>
278                         <span class="on" style="left:61%;width:6%"></span>
279                         <span class="on" style="left:70%;width:13%"></span>
280                         <span class="on" style="left:86%;width:9%"></span>
281                     </div>
282                     <div class="result">
283                         <span class="play" aria-hidden="true">?</span>
284                         <span class="result-tx"><b>Highlight reel · 44 rallies · 18 min</b><em>dead
time removed · ready to download</em></span>
285                         <span class="dl" aria-hidden="true">?</span>
286                     </div>
287                 </div>
288             </div>

```

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289     </div>
290 </div>
291 </section>
292
293 <section id="how">
294     <div class="wrap">
295         <div class="sec-head">
296             <div class="kicker">What it does today</div>
297             <h2>A full match, turned into rallies you can search</h2>
298             <p>Upload a whole session or match. KhelSutra does the rest ? no editing skill
needed.</p>
299         </div>
300         <div class="grid g4">
301             <div class="card">
302                 <div class="ic" aria-hidden="true">?</div>
303                 <h3>Every rally, indexed</h3>
304                 <p>It finds each rally ? start, end and length ? and turns the whole match into
a searchable list. Long full-session videos are fine.</p>
305             </div>
306             <div class="card">
307                 <div class="ic" aria-hidden="true">?</div>
308                 <h3>Dead time removed</h3>
309                 <p>Knock-ups, walks, picking up the shuttle, the gaps between points ? stripped
out. What's left is only real play.</p>
310             </div>
311             <div class="card">
312                 <div class="ic" aria-hidden="true">?</div>
313                 <h3>Process once, reuse</h3>
314                 <p>Come back later to watch or download your reel. Re-running a video replaces
its earlier result rather than piling up copies.</p>
315             </div>
316             <div class="card">
317                 <div class="ic" aria-hidden="true">?</div>
318                 <h3>Stitched reel + summary</h3>
319                 <p>Your selection becomes one MP4 to download and share ? with a play-time
summary and a small ?Generated by Khelsutra.guru? mark.</p>
320             </div>
321         </div>
322     </div>
323 </section>
324
325 <section id="record" style="background:#EFF4EC">
326     <div class="wrap">
327         <div class="sec-head">
328             <div class="kicker">Record it right</div>
329             <h2>Five minutes of setup ? reliably great reels</h2>
330             <p>Highlight quality starts with the recording. Set the camera up once like this
and KhelSutra has everything it needs to find the rallies.</p>
331         </div>
332
333         <div style="max-width:780px;margin:0 auto 1.6rem;display:flex;gap:.8rem;align-
items:flex-start;background:#fff;border:1px solid var(--line);border-left:4px solid
var(--green);border-radius:14px;padding:1rem 1.2rem">
334             <span aria-hidden="true" style="color:var(--green-d);font-size:1.4rem;line-
height:1">?</span>
335             <p style="margin:0;color:var(--tx)"><b>The golden rule: one match, one court, in
full frame ? and don't move the camera.</b> <span style="color:var(--mut)">A single court in
frame gives the best results today; busy multi-court footage is the hard case we're still
sharpening.</span></p>
336         </div>
337
338     <div class="grid g4">
339         <div class="card">
340             <div class="ic" aria-hidden="true">?</div>
341             <h3>Place the camera</h3>

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342         <p>Behind <b>one baseline</b>, elevated (a high tripod or a balcony), tilted
down ~30° ? the <b>whole court</b> in frame, players never cropped.</p>
343     </div>
344     <div class="card">
345         <div class="ic" aria-hidden="true">?</div>
346         <h3>Settings</h3>
347         <p><b>1080p</b>, landscape, <b>30 fps</b> for badminton (60 fps for table
tennis). <b>Audio ON</b> ? the shuttle's ?thwack? helps find rallies.</p>
348     </div>
349     <div class="card">
350         <div class="ic" aria-hidden="true">?</div>
351         <h3>Keep it still</h3>
352         <p>Fixed tripod or mount. <b>No panning, no zooming</b> during play ? a steady,
full-court view is what the tool is tuned for.</p>
353     </div>
354     <div class="card">
355         <div class="ic" aria-hidden="true">?</div>
356         <h3>Record & upload</h3>
357         <p>Even lighting, then record the <b>whole session</b> and upload it as-is ?
KhelSutra finds the rallies and does the cutting.</p>
358     </div>
359 </div>
360
361     <div style="text-align:center;margin-top:1.6rem">
362         <a class="btn btn-dark" href="/capture">Full 1-page capture checklist ?</a>
363     </div>
364 </div>
365 </section>
366
367 <section id="diff" style="padding-top:0">
368     <div class="wrap">
369         <div class="split">
370             <div>
371                 <div class="kicker" style="font-size:.82rem;font-weight:700;letter-
spacing:.08em;text-transform:uppercase;color:var(--green-d)">The hard problem we crack</div>
372                 <h2 style="margin-top:.5rem">Most academy halls run several courts at once</h2>
373                 <p style="color:var(--mut);font-size:1.05rem">Academies already record a lot of
video ? and then almost never watch it, because cutting it into useful clips by hand is tedious.
The good rallies stay buried.</p>
374                 <p style="color:var(--mut);font-size:1.05rem">A single recording of a busy,
multi-court hall is exactly where ordinary video tools choke. Pulling the <b
style="color:var(--tx)">real</b> rallies out of busy, multi-court footage is the hard problem
KhelSutra is built for. It already works very well on clean, single-court recordings ? and the
multi-court case is what we're sharpening now.</p>
375             </div>
376             <div class="panel">
377                 <span class="pill">Capture tip</span>
378                 <h2 style="font-size:1.4rem">Best results: a steady, full-court shot</h2>
379                 <p>Camera behind the court, slightly elevated, whole court in frame. Keep it
still ? a fixed tripod beats handheld every time. Shoot landscape, record the whole session,
upload as-is. KhelSutra does the cutting.</p>
380             </div>
381         </div>
382     </div>
383 </section>
384
385 <section id="sharper" style="background:#EFF4EC">
386     <div class="wrap">
387         <div class="sec-head">
388             <div class="kicker">What makes it different</div>
389             <h2>It gets sharper from real academy footage</h2>
390             <p>The footage real academies contribute ? especially the busy multi-court halls ?
is what makes KhelSutra better over time. Early users aren't just customers; they're co-
builders.</p>
391         </div>

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392     <div class="grid g3">
393         <div class="card">
394             <div class="ic" aria-hidden="true">?</div>
395             <h3>Improves with the library</h3>
396             <p>When you choose to share footage and coaches confirm what it found, that
growing labelled library is what sharpens future versions ? rather than staying frozen.</p>
397         </div>
398         <div class="card">
399             <div class="ic" aria-hidden="true">?</div>
400             <h3>Your data stays yours</h3>
401             <p>We never use your video to improve the tool without your clear, explicit opt-
in. We never publish your footage. You decide what, if anything, you share.</p>
402         </div>
403         <div class="card">
404             <div class="ic" aria-hidden="true">?</div>
405             <h3>Minors are protected</h3>
406             <p>Footage involving minors is never used to improve the tool and is never
published ? full stop.</p>
407         </div>
408     </div>
409     <div class="honest">
410         <b>Read this honestly:</b>
411         <p>Today the tool gives you rally detection, reels, and summaries. It does not
retrain itself as you use it, and there's no per-person tracking. "Gets sharper over time" is
the model we're building toward, powered by contributed footage ? not a switch that's already
on.</p>
412     </div>
413 </div>
414 </section>
415
416 <section id="ask">
417     <div class="wrap">
418         <div class="sec-head">
419             <div class="kicker">Coming next to the alpha</div>
420             <h2>Soon: ask for the exact reel you want</h2>
421             <p>Your match is already a clean index of rallies ? so the natural next step is to
let you build any reel just by describing it, in plain English. This is rolling out to alpha
users.</p>
422         </div>
423         <div class="grid g3 ask-grid">
424             <div class="card askc">
425                 <div class="q"><i>?</i>Every rally over 10 seconds?</div>
426                 <div class="a"><i>?</i> 6 rallies · a 1:32 reel of the long, end-to-end
battles</div>
427             </div>
428             <div class="card askc">
429                 <div class="q"><i>?</i>Just the third game, longest first?</div>
430                 <div class="a"><i>?</i> 12 rallies, reordered by length ? that game only</div>
431             </div>
432             <div class="card askc">
433                 <div class="q"><i>?</i>The 8 best rallies from today?</div>
434                 <div class="a"><i>?</i> an 8-clip highlight from your latest session</div>
435             </div>
436         </div>
437         <p class="ask-note"><b style="color:var(--tx)">Rolling out now.</b> Targeted asks ?
by rally length, count, order, or which game ? come first; shot-type and player queries follow
as the deeper analysis arrives.</p>
438     </div>
439 </section>
440
441 <section id="roadmap">
442     <div class="wrap">
443         <div class="sec-head">
444             <div class="kicker">Where it's going</div>
445             <h2>Toward an AI assistant coach</h2>

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446         <p>Fed by the growing video library, KhelSutra is heading from "finds rallies" to
a genuine training companion. This is the direction ? not today's buttons.</p>
447     </div>
448     <div class="road">
449         <div class="step"><div class="n">?</div><h3>Shot detection</h3><p>Smashes, drops,
net shots, serves and clears within each rally.</p></div>
450         <div class="step"><div class="n">?</div><h3>Deeper analysis</h3><p>Per-rally and
per-match patterns ? what's working and what isn't.</p></div>
451         <div class="step"><div class="n">?</div><h3>Per-match history</h3><p>Track
progress over weeks and months, not just one clip.</p></div>
452         <div class="step"><div class="n">?</div><h3>Assistant coach</h3><p>Surfaces what
to work on next ? supports the real coach, never replaces.</p></div>
453         <div class="step"><div class="n">?</div><h3>More sports & languages</h3><p>Table
tennis next; Kannada and Hindi alongside English.</p></div>
454     </div>
455 </div>
456 </section>
457
458 <section id="not" style="padding-top:0">
459     <div class="wrap">
460         <div class="sec-head">
461             <div class="kicker">So expectations stay honest</div>
462             <h2>What KhelSutra is not</h2>
463         </div>
464         <ul class="nots">
465             <li><i aria-hidden="true">?</i><span>An early, invite-only trial ? useful, but
still improving. You'll hit rough edges.</span></li>
466             <li><i aria-hidden="true">?</i><span>Not a line-calling, scoring, or umpiring
system.</span></li>
467             <li><i aria-hidden="true">?</i><span>Not player tracking or
identification.</span></li>
468             <li><i aria-hidden="true">?</i><span>Not a replacement for a coach ? a tool that
saves time and adds insight.</span></li>
469             <li><i aria-hidden="true">?</i><span>Shot detection and deeper analytics are on
the roadmap, not available today.</span></li>
470             <li><i aria-hidden="true">?</i><span>Pricing isn't set ? today's invited use is
about getting it right.</span></li>
471         </ul>
472     </div>
473 </section>
474
475 <section id="faq" style="background:#EFF4EC">
476     <div class="wrap">
477         <div class="sec-head">
478             <div class="kicker">Quick answers</div>
479             <h2>Common questions</h2>
480         </div>
481         <div style="margin-top:1.6rem">
482             <details><summary>How accurate is it?</summary><p>Very good on a clean, single-
court recording. A busy multi-court hall is harder ? and is exactly what we're improving. The
best answer is to show you a sample reel of your own court rather than quote a
number.</p></details>
483             <details><summary>Can I choose which rallies go in the reel?</summary><p>Today
your reel includes the rallies KhelSutra found, with the dead time removed. Choosing by length,
count, order or which game ? just by asking in plain English ? is rolling out to alpha users
next; shot-type and player queries follow as the deeper analysis arrives.</p></details>
484             <details><summary>Do I need a special camera or hardware?</summary><p>No. A phone
or GoPro on a tripod, behind the court, whole court in frame. We provide a one-page capture
checklist at setup.</p></details>
485             <details><summary>Is my video private? What happens to it?</summary><p>We treat
your footage as private: we never publish it, and we never use it to improve the tool without
your explicit opt-in. These are large videos that need real processing, so your video is handled
securely on our infrastructure to make your reel. Footage involving minors is never used and
never published.</p></details>
486             <details><summary>Does it work for doubles and tournaments?</summary><p>It finds
```

rallies in doubles too. Tournament footage varies ? a clean, fixed, full-court recording works well; crowded multi-court tournament halls are the hard case we're specifically working on.

Which sports?

Badminton today. Table tennis and other racket sports are on the roadmap.

How do I join?

It's an invite-only private trial. Just [request access](#join) using the form ? it takes a minute, you'll see a real sample reel right away, and we'll be in touch about recording your own court.

`<section id="join" style="padding-bottom:90px">`

`<div class="wrap">`

`<div class="sec-head" style="text-align:center;max-width:none">`

`<div class="kicker">Request access</div>`

`<h2>Get into the KhelSutra alpha</h2>`

`<p>Invite-only. Tell us a little about your setup ? it helps us get you the best results, and shapes what we build next.</p>`

`<p style="font-weight:600;color:var(--green-d);margin-top:.6rem"><span aria-hidden="true">?</span> Sign up to watch the tool in action ? and a 30-second sample reel it made ? the moment you're in.</p>`

`</div>`

`<form id="af" class="af" novalidate>`

`<div class="row">`

`<label>I'm a?</label>`

`<div class="seg">`

`<label><input type="radio" name="segment" value="academy"`

`required><span>Coaching academy / club</span></label>`

`<label><input type="radio" name="segment" value="individual"><span>Individual player</span></label>`

`</div>`

`</div>`

`<div class="row">`

`<label for="af-name">Name</label>`

`<input id="af-name" name="name" type="text" autocomplete="name" required>`

`</div>`

`<div class="row">`

`<label for="af-email">Email <span class="hint">(so we can reach you)</span></label>`

`<input id="af-email" name="email" type="email" autocomplete="email" required>`

`</div>`

`<div class="row">`

`<label for="af-city">City <span class="hint">(optional)</span></label>`

`<input id="af-city" name="city" type="text">`

`</div>`

`<div class="row acad">`

`<label for="af-org">Academy / club name <span class="hint">(optional)</span></label>`

`<input id="af-org" name="org_name" type="text">`

`</div>`

`<div class="row acad">`

`<label for="af-courts">How many courts do you record? <span class="hint">(optional)</span></label>`

`<select id="af-courts" name="courts">`

`<option value="">Select?</option>`

`<option value="1">1 court</option>`

`<option value="2-4">2?4 courts</option>`

`<option value="5+">5+ courts</option>`

`</select>`

`</div>`

```

538         <div class="row">
539             <label>Do you have a GPU machine that could process video locally?</label>
540             <div class="opts">
541                 <label><input type="radio" name="has_gpu" value="yes">Yes ? a gaming / GPU PC
or laptop</label>
542                 <label><input type="radio" name="has_gpu" value="unsure">Not sure</label>
543                 <label><input type="radio" name="has_gpu" value="no">No</label>
544                 <label><input type="radio" name="has_gpu" value="prefer-cloud">I'd prefer it
runs in the cloud</label>
545             </div>
546         </div>
547
548         <div class="row">
549             <label>Where do your recordings live / how would you share them? <span
class="hint">(pick any)</span></label>
550             <div class="opts">
551                 <label><input type="checkbox" name="storage" value="gdrive">Google
Drive</label>
552                 <label><input type="checkbox" name="storage" value="onedrive">OneDrive / MS
365</label>
553                 <label><input type="checkbox" name="storage" value="dropbox">Dropbox</label>
554                 <label><input type="checkbox" name="storage" value="bucket">Cloud bucket
(S3/GCS)</label>
555                 <label><input type="checkbox" name="storage" value="local">SD card / local
drive</label>
556                 <label><input type="checkbox" name="storage" value="unsure">Not sure</label>
557             </div>
558         </div>
559
560         <div class="row">
561             <label for="af-msg">Anything else? <span class="hint">(how you record, what
you'd want?)</span></label>
562             <textarea id="af-msg" name="message"></textarea>
563         </div>
564
565         <div class="row consent">
566             <input id="af-consent" name="consent" type="checkbox" value="yes">
567             <label for="af-consent" style="margin:0;font-weight:400;color:var(--mut)">Okay
to email me about the KhelSutra alpha.</label>
568         </div>
569
570         <input class="hp" type="text" name="company" tabindex="-1" autocomplete="off"
aria-hidden="true">
571
572         <button class="btn btn-primary" type="submit" style="width:100%;justify-
content:center">Request access</button>
573         <p class="hint" style="text-align:center;margin:.85rem 0 0">We never publish your
footage; footage involving minors is never used to improve the tool.</p>
574     </form>
575
576     <p id="af-done" class="form-done" hidden>Thanks ? we've got it. We'll be in touch at
the email you gave. ?<br><span style="font-weight:400;color:var(--mut)">While you're here: see
the tool in action, and a real 30-second reel it made.</span></p>
577     <div id="reel-reveal" hidden></div>
578 </div>
579 </section>
580
581 <footer>
582     <div class="wrap foot">
583         <div>
584             <div class="brand" style="font-size:1.05rem">Khel<span class="play" aria-
hidden="true"></span>Sutra</div>
585             <div class="stamp" style="margin-top:.4rem">Automatic badminton highlights .
Bengaluru, India</div>
586         </div>

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587         <div style="text-align:right">
588             <div><a href="/own-model">Own your model</a> . <a href="/capture">Capture
checklist</a> . <a href="mailto:hello@khelsutra.guru">hello@khelsutra.guru</a> . <a
href="https://khelsutra.guru">khelsutra.guru</a></div>
589             <div style="margin-top:.4rem">Academy flyer (print): <a href="/flyer"
hreflang="en">English</a> . <a href="/flyer-kn" hreflang="kn" lang="kn">????</a> . <a
href="/flyer-hi" hreflang="hi" lang="hi">????</a></div>
590             <div class="stamp" style="margin-top:.4rem">© 2026 KhelSutra . Invite-only alpha .
Reels carry a "Generated by Khelsutra.guru" mark</div>
591         </div>
592     </div>
593 </footer>
594
595     <!-- Cloudflare Web Analytics: paste your token from Cloudflare dashboard ? Analytics ?
Web Analytics ? Add a site.
596         (Or skip this snippet entirely and enable "Automatic setup" once the domain is
proxied through Cloudflare.) -->
597     <script defer src="https://static.cloudflareinsights.com/beacon.min.js" data-cf-
beacon="{\"token\": \"REPLACE_WITH_CF_TOKEN\"}"></script>
598
599     <script>
600     (function(){
601         var f = document.getElementById('af');
602         if(!f) return;
603
604         var REEL_VIDEO = 'ajTCZ7pJW_o', REEL_SECS = 30;
605         function loadYTApi(cb){
606             if(window.YT && window.YT.Player) return cb();
607             var prev = window.onYouTubeIframeAPIReady;
608             window.onYouTubeIframeAPIReady = function(){ if(typeof prev === 'function'){ prev();
} cb(); };
609             if(!document.getElementById('yt-iframe-api')){
610                 var s = document.createElement('script');
611                 s.id = 'yt-iframe-api'; s.src = 'https://www.youtube.com/iframe_api';
612                 document.head.appendChild(s);
613             }
614         }
615         function revealAfterSignup(){
616             var host = document.getElementById('reel-reveal');
617             if(!host) return;
618             host.hidden = false;
619             // (1) the TOOL in action ? a self-hosted, FOOTAGE-FREE screencast (branded reel
placeholder),
620             //         so it's safe to autoplay openly; (2) a real reel preview (gated nocookie
YouTube). BOTH
621             //         stay visible together; the auto-close PAUSES while the reel is playing (so a
viewer is
622             //         never cut off mid-watch) and a manual ? closes it on demand.
623             host.innerHTML =
624                 '<div class="reveal-bar"><button type="button" id="reel-close" class="reveal-x"
aria-label="Close the videos">? Close</button></div>'
625                 + '<p class="reveal-h">1 . The tool in action</p>'
626                 + '<video class="tool-vid" src="/video/tool-cuj.mp4" autoplay muted loop
playsinline controls preload="metadata"></video>'
627                 + '<p class="reveal-h">2 . A real reel it made</p>'
628                 + '<div class="reelwrap"><div id="reel-player"></div></div>'
629                 + '<p class="stamp" style="text-align:center;margin-top:.6rem">Generated by
Khelsutra.guru . auto-closes <span id="reel-count">' + REEL_SECS + '</span>s after the reel
stops ? or close it anytime</p>';
630             var n = REEL_SECS, el = document.getElementById('reel-count'), player = null, hold =
false, closed = false;
631             function close(){
632                 if(closed) return; closed = true;
633                 clearInterval(iv);
634                 try{ if(player){ player.destroy(); } }catch(_){ }

```

```

635     host.innerHTML = ''; host.hidden = true;
636 }
637 var iv = setInterval(function(){
638     if(hold) return; // the reel is playing ? hold the timer (no
mid-watch cut-off)
639     n--; if(el){ el.textContent = n; }
640     if(n <= 0){ close(); }
641 }, 1000);
642 var x = document.getElementById('reel-close'); if(x){ x.addEventListener('click',
close); }
643 loadYTApi(function(){
644     player = new YT.Player('reel-player', {
645         width: '100%', height: '100%', videoId: REEL_VIDEO,
646         host: 'https://www.youtube-nocookie.com',
647         playerVars: { autoplay: 1, rel: 0, modestbranding: 1, playsinline: 1 },
648         events: {
649             onReady: function(e){
650                 var dur = Math.floor(e.target.getDuration() || 0);
651                 var maxStart = Math.max(0, dur - REEL_SECS); // start so a full
~30s window plays before the end
652                 var start = maxStart > 0 ? Math.floor(Math.random() * (maxStart + 1)) : 0;
653                 host.setAttribute('data-reel-start', String(start)); // a different chunk
each sign-up
654                 e.target.seekTo(start, true);
655                 e.target.playVideo();
656             },
657             onStateChange: function(e){ hold = (e.data === 1); } // 1 = PLAYING ? hold
the auto-close
658         }
659     });
660 });
661 }
662
663 f.addEventListener('change', function(e){
664     if(e.target.name === 'segment') f.classList.toggle('is-academy', e.target.value ===
'academy');
665 });
666 function mailtoFallback(p){
667     var lines = [
668         'Segment: ' + (p.segment||''),
669         'Name: ' + (p.name||''),
670         'Email: ' + (p.email||''),
671         'City: ' + (p.city||''),
672         p.segment === 'academy' ? 'Academy: ' + (p.org_name||'') + ' (' + (p.courts||'')
+ ' courts)' : null,
673         'GPU: ' + (p.has_gpu||''),
674         'Storage: ' + ((p.storage||[]).join(', ')),
675         'Message: ' + (p.message||''),
676         'Consent: ' + (p.consent ? 'yes' : 'no')
677     ].filter(Boolean).join('\n');
678     return 'mailto:hello@khelsutra.guru?subject=' + encodeURIComponent('KhelSutra alpha
access') +
679         '&body=' + encodeURIComponent('Hi KhelSutra team, I would like to try the
alpha.\n\n' + lines);
680 }
681 f.addEventListener('submit', async function(e){
682     e.preventDefault();
683     if(f.company && f.company.value) return; // honeypot: silently drop bots
684     if(!f.checkValidity()){ f.reportValidity(); return; }
685     // Immediate, obvious "submitting" state ? the Worker + D1 + email round-trip can
take a beat,
686     // and a static button looked frozen.
687     var btn = f.querySelector('button[type="submit"]');
688     var btnHTML = btn ? btn.innerHTML : '';
689     if(btn){ btn.disabled = true; btn.setAttribute('aria-busy','true'); btn.innerHTML =

```

```

'<span class="spin" aria-hidden="true"></span> Sending?'; }
690     var fd = new FormData(f);
691     var p = {
692         segment: fd.get('segment'), name: fd.get('name'), email: fd.get('email'),
693         city: fd.get('city'), org_name: fd.get('org_name'), courts: fd.get('courts'),
694         has_gpu: fd.get('has_gpu'), storage: fd.getAll('storage'),
695         message: fd.get('message'), consent: !!fd.get('consent')
696     };
697     var ok = false;
698     try {
699         var r = await fetch('/api/request-access', {method:'POST', headers: {'content-
type':'application/json'}, body: JSON.stringify(p)});
700         ok = r.ok;
701     } catch(_){ ok = false; }
702     if(ok){
703         f.hidden = true;
704         document.getElementById('af-done').hidden = false;
705         revealAfterSignup();
706     }
707     else {
708         // Restore the button so the user isn't left with a frozen "Sending?", then fall
back to email.
709         if(btn){ btn.disabled = false; btn.removeAttribute('aria-busy'); btn.innerHTML =
btnHTML; }
710         window.location.href = mailtoFallback(p); // backend not live yet ? prefilled
email
711     }
712     });
713     })();
714 </script>
715
716 </body>
717 </html>
718

```

12. user (2026-06-21T14:34:18.682Z)

No files found

13. user (2026-06-21T14:34:19.008Z)

No files found

14. assistant (2026-06-21T14:34:24.635Z)

The glob for `web/src/i18n/locales` returned nothing, which is odd since index.ts imports them. Let me check the actual structure and the site src dir and capture/own-model HTML, plus check for the existing site test files and how the SPA tests negotiation. Let me also look at the engine I18N_PLAN.md P1 section.

15. user (2026-06-21T14:34:28.315Z)

=== web/src/i18n tree ===

```

web/src/i18n:
hygiene.test.ts
i18n.test.tsx
index.ts
locales

```

```

web/src/i18n/locales:
en
hi
kn

```

```
web/src/i18n/locales/en:
common.json
```

```
web/src/i18n/locales/hi:
common.json
```

```
web/src/i18n/locales/kn:
common.json
```

```
=== site/src ===
assets.test.js
deploy-config.test.js
index.js
index.test.js
notify.js
server.test.js
store.js
ux.dom.test.js
=== site tests ===
ls: cannot access 'site/*.test.js': No such file or directory
site/src/assets.test.js
site/src/deploy-config.test.js
site/src/index.test.js
site/src/server.test.js
site/src/ux.dom.test.js
=== web tests dir ===
web/src/api/client.test.ts
web/src/components/IngestPanel.test.tsx
web/src/components/QueryBar.examples.test.ts
web/src/hooks/useIngestJob.test.ts
web/src/i18n/hygiene.test.ts
web/src/i18n/hygiene.test.ts
web/src/i18n/i18n.test.tsx
web/src/i18n/i18n.test.tsx
web/src/lib/errors.test.ts
web/src/lib/floor.test.ts
=== package.json (site) ===
{
  "name": "khelsutra-site",
  "private": true,
  "type": "module",
  "scripts": {
    "test": "node --test",
    "test:cf": "node --test src/index.test.js",
    "test:vm": "node --test src/server.test.js",
    "test:deploy": "node --test src/deploy-config.test.js src/assets.test.js",
    "deploy": "wrangler deploy",
    "dev": "wrangler dev",
    "og": "python make_og.py"
  },
  "devDependencies": {
    "jsdom": "^29.1.1"
  }
}
```

16. assistant (2026-06-21T14:34:35.691Z)

Now let me read the key reference files: the SPA i18n test (to mirror negotiation parity in the site test), the site server.test.js and ux.dom.test.js (to match test conventions), the capture/own-model HTML, and the engine I18N_PLAN.md P1 section.

17. user (2026-06-21T14:34:36.599Z)

```
1      // @vitest-environment jsdom
2      //
3      // The SPA had NO render tests before this (verified) ? this file adds the jsdom safety
```

```

net for the
4      // i18n scaffold (tracker B1): real copy renders from the catalog, the switcher persists
the choice,
5      // missing-locale falls back to English, and ICU plurals format. It mounts the real <App
/>.
6      import { describe, it, expect, beforeEach, afterEach, vi } from "vitest";
7      import { render, screen, cleanup, fireEvent } from "@testing-library/react";
8      import i18n, { LANG_STORAGE_KEY } from "../index";
9      import App from "../App";
10
11      beforeEach(async () => {
12          // No engine in the test env: stub fetch so App's getConfig()/listRallies effects
reject quietly
13          // (App handles this gracefully ? cfg stays null) instead of hitting the network.
14          vi.stubGlobal("fetch", vi.fn(() => Promise.reject(new Error("no engine in test"))));
15          window.localStorage.clear();
16          await i18n.changeLanguage("en"); // also guarantees init completed before the first
render
17      });
18
19      afterEach(() => {
20          cleanup();
21          vi.unstubAllGlobals();
22      });
23
24      describe("i18n scaffold (B1)", () => {
25          it("renders user-facing copy from the en catalog (not raw keys)", () => {
26              render(<App />);
27              expect(screen.getByRole("heading", { level: 1 })).textContent.toBe("Pick the rallies
for your reel");
28              expect(screen.getByRole("button", { name: "Select all" })).toBeTruthy();
29          });
30
31          it("B2: the extracted component strings render (no leftover raw keys across the
page)", () => {
32              const { container } = render(<App />);
33              expect(screen.getByRole("button", { name: "Apply" })).toBeTruthy(); // QueryBar
34              expect(screen.getByRole("button", { name: /Build reel/ })).toBeTruthy(); //
SelectionFooter
35              expect(container.textContent).toContain("rally length"); // <Trans> hero intro +
query support
36              expect(container.textContent).toMatch(/\d+ rallies selected · \d+:\d+ total/); //
ICU footer summary
37              expect(container.textContent).toMatch(/Rally \d+/); // RallyGrid card
38              // No untranslated catalog keys leaked into the DOM.
39              expect(container.textContent).not.toMatch(/\b(app|grid|footer|query|mode|result|buil
d)\.[a-zA-Z]+\b/);
40          });
41
42          it("offers a language switcher and persists the choice (persisted pref wins on next
visit)", () => {
43              render(<App />);
44              const knBtn = screen.getByRole("button", { name: "?????" });
45              fireEvent.click(knBtn);
46              expect(i18n.language).toBe("kn");
47              expect(window.localStorage.getItem(LANG_STORAGE_KEY)).toBe("kn");
48              expect(knBtn.getAttribute("aria-pressed")).toBe("true"); // highlights the chosen
locale
49              // Switching actually re-renders the page into Kannada (the kn catalog, B5).
50              expect(screen.getByRole("heading", { level: 1 })).textContent.toBe("????? ?????????
????????????? ??????");
51          });
52
53          it("B5: switching to Kannada renders the kn catalog (Kannada copy, not English)",
async () => {

```

```

54     await i18n.changeLanguage("kn");
55     render(<App />);
56     expect(screen.getByRole("heading", { level: 1 }).textContent).toBe("????? ??????????
????????????? ?????");
57     expect(screen.getByRole("button", { name: "?????????" })).toBeTruthy(); // Apply ?
Kannada
58     expect(screen.getByRole("button", { name: "????????? ??????????" })).toBeTruthy(); //
Select all ? Kannada
59     expect(document.documentElement.lang).toBe("kn"); // <html lang> tracks the active
locale (a11y)
60     });
61
62     it("B6: switching to Hindi renders the hi catalog (extensibility proof ? just a
catalog)", async () => {
63         await i18n.changeLanguage("hi");
64         render(<App />);
65         expect(screen.getByRole("heading", { level: 1 }).textContent).toBe("???? ??? ?? ??
????????? ?????");
66         expect(screen.getByRole("button", { name: "???? ??????" })).toBeTruthy(); // Apply ?
Hindi
67         expect(document.documentElement.lang).toBe("hi");
68     });
69
70     it("a locale with no catalog falls back to English (never a raw key)", async () => {
71         await i18n.changeLanguage("ta"); // ta has no catalog
72         render(<App />);
73         expect(screen.getByRole("heading", { level: 1 }).textContent).toBe("Pick the rallies
for your reel");
74     });
75
76     it("formats ICU plurals from the catalog", () => {
77         expect(i18n.t("selection.count", { count: 1 })).toBe("1 rally selected");
78         expect(i18n.t("selection.count", { count: 3 })).toBe("3 rallies selected");
79     });
80 });
81
82 // The owner's bar (#41): a user whose BROWSER/OS language is Kannada/Hindi must auto-
get the
83 // respective version. These prove the negotiation NOW at the detection level (B1, en-
only); the
84 // kn/hi CONTENT assertions ride the catalogs in B5/B6. See FRIENDLY_AND_MULTILINGUAL.md
$6.5 (N1?N9).
85 describe("browser-language negotiation (B1: N1/N3?N7)", () => {
86     const detectBases = () => {
87         const d = i18n.services.languageDetector.detect();
88         return (Array.isArray(d) ? d : [d]).filter(Boolean).map((l) =>
String(l).split("-")[0]);
89     };
90
91     it("N1/N5: a Kannada browser (kn-IN) is auto-detected as kn for a first-time visitor",
() => {
92         window.localStorage.clear(); // no persisted preference yet
93         vi.stubGlobal("navigator", { language: "kn-IN", languages: ["kn-IN", "en-IN"] });
94         expect(detectBases()).toContain("kn"); // the kn-IN region subtag normalizes to base
kn
95     });
96
97     it("N2: a Hindi browser (hi-IN) is auto-detected as hi (now a supported locale)", ()
=> {
98         window.localStorage.clear();
99         vi.stubGlobal("navigator", { language: "hi-IN", languages: ["hi-IN", "en-IN"] });
100         expect(detectBases()).toContain("hi");
101     });
102
103     it("N4: a persisted preference beats the browser language (returning user keeps their

```

```

choice)", () => {
104     window.localStorage.setItem(LANG_STORAGE_KEY, "kn");
105     vi.stubGlobal("navigator", { language: "en-US", languages: ["en-US"] });
106     expect(detectBases()[0]).toBe("kn");
107 });
108
109     it("N3: persisted pref and ?lang are negotiated BEFORE the browser language", () => {
110         const order = i18n.options.detection?.order ?? [];
111         expect(order.indexOf("localStorage")).toBeGreaterThanOrEqual(0);
112         expect(order.indexOf("localStorage")).toBeLessThan(order.indexOf("navigator"));
113         expect(order.indexOf("querystring")).toBeLessThan(order.indexOf("navigator"));
114     });
115
116     it("N6: an unsupported browser language (Tamil) serves English content ? never a
locale we can't render", async () => {
117         window.localStorage.clear();
118         await i18n.changeLanguage("ta"); // ta is not in SUPPORTED_LOCALES
119         render(<App />);
120         expect(screen.getByRole("heading", { level: 1 }).textContent).toBe("Pick the rallies
for your reel");
121     });
122
123     it("N7: a key with no catalog entry falls back to English (no raw key)", async () => {
124         await i18n.changeLanguage("ta"); // ta has no catalog ? keys resolve via
fallbackLng=en
125         expect(i18n.t("app.title")).toBe("Pick the rallies for your reel");
126         expect(i18n.t("bulk.selectAll")).toBe("Select all");
127     });
128 });
129

```

18. user (2026-06-21T14:34:37.161Z)

```

1     // [VM/droplet host harness] Boots the REAL server.js (node:http + node:sqlite) and
2     // round-trips it over HTTP ? the same proof we run on the droplet, automated. Keeps the
3     // repo compatible with BOTH hosts: src/index.test.js covers the Cloudflare (Worker+D1)
4     // path; this covers the Node+SQLite path. Run: node --test src/ (or: npm run test:vm)
5     import { test, before, after } from "node:test";
6     import assert from "node:assert/strict";
7     import { spawn } from "node:child_process";
8     import { createServer } from "node:net";
9     import { mkdtempSync, rmSync } from "node:fs";
10    import { tmpdir } from "node:os";
11    import { join } from "node:path";
12    import { DatabaseSync } from "node:sqlite";
13
14    function freePort() {
15        return new Promise((res) => {
16            const s = createServer();
17            s.listen(0, "127.0.0.1", () => { const p = s.address().port; s.close(() => res(p));
18        });
19    }
20    async function waitUp(url, ms = 10000) {
21        const end = Date.now() + ms;
22        while (Date.now() < end) {
23            try { await fetch(url); return; } catch { await new Promise((r) => setTimeout(r,
150)); }
24        }
25        throw new Error("server.js did not start listening in time");
26    }
27
28    let proc, base, dbPath, dir;
29
30    before(async () => {

```

```

31     const port = await freePort();
32     dir = mkdtempSync(join(tmpdir(), "ks-vm-"));
33     dbPath = join(dir, "access.db");
34     base = `http://127.0.0.1:${port}`;
35     const serverPath = join(import.meta.dirname, "..", "server.js");
36     proc = spawn(process.execPath, [serverPath], {
37       env: { ...process.env, PORT: String(port), DB_PATH: dbPath },
38       stdio: "ignore",
39     });
40     await waitUp(`${base}/`);
41   });
42
43   after(() => {
44     proc?.kill();
45     try { rmSync(dir, { recursive: true, force: true }); } catch {}
46   });
47
48   test("[VM host] server.js serves the form", async () => {
49     const html = await (await fetch(`${base}/`)).text();
50     assert.match(html, /id="af"/);
51     assert.match(html, /Get into the KhelSutra alpha/);
52   });
53
54   test("[VM host] serves /capture and /og.png", async () => {
55     assert.equal((await fetch(`${base}/capture`)).status, 200);
56     assert.equal((await fetch(`${base}/og.png`)).status, 200);
57   });
58
59   test("[VM host] serves the tool screencast as video/mp4 (plays in the post-signup reveal)", async () => {
60     const res = await fetch(`${base}/video/tool-cuj.mp4`);
61     assert.equal(res.status, 200);
62     assert.equal(res.headers.get("content-type"), "video/mp4"); // not octet-stream ?
63     <video> can play it
64   });
65
66   test("[VM host] serves /own-model", async () => {
67     const res = await fetch(`${base}/own-model`);
68     assert.equal(res.status, 200);
69     assert.match(await res.text(), /Train a model that's yours/);
70   });
71
72   test("[VM host] serves the academy flyers (/flyer, /flyer-kn, /flyer-hi) as clean URLs",
73   async () => {
74     // Cloudflare maps these via asset html-handling; the VM host needs explicit routes so
75     the
76     // footer links work on BOTH hosts (parity).
77     for (const path of ["/flyer", "/flyer-kn", "/flyer-hi"]) {
78       const res = await fetch(`${base}${path}`);
79       assert.equal(res.status, 200, `${path} should serve 200 on the VM host`);
80       assert.match(res.headers.get("content-type"), /text\/html/, `${path} should serve
81       HTML`);
82       assert.match(await res.text(), /<html/i, `${path} should be an HTML page`);
83     }
84   });
85
86   test("[VM host] POST /api/request-access persists to SQLite", async () => {
87     const res = await fetch(`${base}/api/request-access`, {
88       method: "POST",
89       headers: { "content-type": "application/json" },
90       body: JSON.stringify({ segment: "academy", email: "harness@khelesutra.guru", has_gpu:
91       "yes", storage: ["gdrive", "bucket"], consent: true }),
92     });
93     assert.equal(res.status, 200);
94     assert.deepEqual(await res.json(), { ok: true });

```

```

90
91     const db = new DatabaseSync(dbPath);
92     const row = db.prepare("SELECT email, segment, has_gpu, storage, consent FROM
access_requests").get();
93     assert.equal(row.email, "harness@khelsutra.guru");
94     assert.equal(row.segment, "academy");
95     assert.equal(row.has_gpu, "yes");
96     assert.equal(row.storage, "gdrive,bucket");
97     assert.equal(row.consent, 1);
98   });
99
100   test("[VM host] honeypot + bad email are rejected without persisting", async () => {
101     const bot = await fetch(`${base}/api/request-access`, {
102       method: "POST", headers: { "content-type": "application/json" },
103       body: JSON.stringify({ email: "x@y.co", company: "spam" }),
104     });
105     assert.equal(bot.status, 200); // dropped silently
106     const bad = await fetch(`${base}/api/request-access`, {
107       method: "POST", headers: { "content-type": "application/json" },
108       body: JSON.stringify({ email: "nope" }),
109     });
110     assert.equal(bad.status, 400);
111
112     const db = new DatabaseSync(dbPath);
113     const { n } = db.prepare("SELECT COUNT(*) AS n FROM access_requests").get();
114     assert.equal(n, 1); // only the valid submission above
115   });
116
117   test("[VM host] GET /api/request-access -> 405", async () => {
118     assert.equal((await fetch(`${base}/api/request-access`)).status, 405);
119   });
120

```

19. user (2026-06-21T14:34:37.561Z)

```

1    // Behavioral regression tests for the two USER-FACING fixes on the landing page, run
against the
2    // REAL index.html in a jsdom DOM (the inline script actually executes). These guard
behaviour, not
3    // just markup: a refactor that breaks the submit feedback or the watchable reveal fails
CI.
4    // 1. submit ? an obvious 'Sending?' state (disabled + aria-busy); restored on
failure.
5    // 2. post-signup reveal ? both videos; a ? closes it; the auto-close HOLDS while the
reel plays.
6    // jsdom is a TEST-ONLY devDependency ? the Worker + VM host runtimes stay dependency-
free.
7    import { test } from "node:test";
8    import assert from "node:assert/strict";
9    import { readFileSync } from "node:fs";
10   import { join } from "node:path";
11   import { JSDOM, VirtualConsole } from "jsdom";
12
13   const HTML = readFileSync(join(import.meta.dirname, "..", "public", "index.html"),
"utf8");
14   const tick = () => new Promise((r) => setTimeout(r, 5)); // let the inline async handler
settle
15
16   // Boot the real page; `fetchImpl` decides how /api/request-access resolves.
window.setInterval is
17   // captured so the test drives the reveal countdown deterministically (no real 1s
timers). A fake
18   // YT.Player records its events so we can simulate the reel playing/pausing.
19   function boot(fetchImpl) {
20     const vc = new VirtualConsole(); // swallow jsdom "Not implemented: navigation" (the

```

```

mailto fallback)
21     const captured = {};
22     const dom = new JSDOM(HTML, {
23         runScripts: "dangerously",
24         url: "https://khelsutra.guru/",
25         virtualConsole: vc,
26         beforeParse(window) {
27             window.fetch = fetchImpl;
28             window.setInterval = (fn) => { captured.tick = fn; return 1; };
29             window.clearInterval = () => { captured.cleared = true; };
30             window.YT = {
31                 Player: function (_id, cfg) {
32                     captured.reel = cfg;
33                     if (cfg.events && cfg.events.onReady) {
34                         cfg.events.onReady({ target: { getDuration: () => 120, seekTo() {}},
playVideo() {} } });
35                     }
36                     return { destroy() { captured.destroyed = true; } };
37                 },
38             };
39         },
40     });
41     const { document } = dom.window;
42     const form = document.getElementById("af");
43     // fill the minimum-valid form
44     form.querySelector('input[name="segment"][value="individual"]').checked = true;
45     document.getElementById("af-name").value = "Demo";
46     document.getElementById("af-email").value = "demo@example.com";
47     const consent = document.getElementById("af-consent");
48     if (consent) consent.checked = true;
49     const submit = () => form.dispatchEvent(new dom.window.Event("submit", { bubbles:
true, cancelable: true }));
50     const btn = () => form.querySelector('button[type="submit"]');
51     return { dom, document, form, submit, btn, captured };
52 }
53
54 test("submit shows an obvious in-flight state (disabled + aria-busy + 'Sending?')",
async () => {
55     // a fetch that never resolves ? the button stays in the in-flight state we can
inspect
56     const { submit, btn } = boot(() => new Promise(() => {}));
57     submit();
58     await tick();
59     const b = btn();
60     assert.equal(b.disabled, true, "button must be disabled while submitting");
61     assert.equal(b.getAttribute("aria-busy"), "true", "aria-busy must be set for assistive
tech");
62     assert.match(b.textContent, /Sending?/, "button must read 'Sending?'");
63     assert.ok(b.querySelector(".spin"), "an in-flight spinner must be shown");
64 });
65
66 test("on failure the button is restored ? never stuck on a frozen 'Sending?'", async ()
=> {
67     const { submit, btn } = boot(async () => ({ ok: false })); // backend down ? mailto
fallback path
68     const before = btn().textContent;
69     submit();
70     await tick();
71     const b = btn();
72     assert.equal(b.disabled, false, "button must be re-enabled after a failed submit");
73     assert.equal(b.getAttribute("aria-busy"), null, "aria-busy must be cleared");
74     assert.equal(b.textContent.trim(), before.trim(), "button label must be restored");
75 });
76
77 test("after sign-up BOTH videos appear and a ? closes the reveal", async () => {

```

```

78     const { document, submit } = boot(async () => ({ ok: true }));
79     submit();
80     await tick();
81     const host = document.getElementById("reel-reveal");
82     assert.equal(host.hidden, false, "reveal should be visible after a successful
submit");
83     assert.ok(host.querySelector("video.tool-vid"), "the tool screencast video must be
present");
84     assert.ok(host.querySelector(".reelwrap #reel-player, .reelwrap iframe"), "the reel
player must be present");
85     const close = host.querySelector("#reel-close");
86     assert.ok(close, "a manual ? close must be present");
87     close.click();
88     assert.equal(host.hidden, true, "? must hide the reveal");
89     assert.equal(host.innerHTML, "", "? must clear the reveal");
90   });
91
92   test("the auto-close HOLDS while the reel is playing (no mid-watch cut-off)", async ()
=> {
93     const { document, submit, captured } = boot(async () => ({ ok: true }));
94     submit();
95     await tick();
96     const count = () => Number(document.getElementById("reel-count").textContent);
97     const start = count();
98     captured.tick();
99     assert.equal(count(), start - 1, "the countdown ticks down normally");
100    // reel starts PLAYING (YT state 1) ? the countdown must FREEZE
101    captured.reel.events.onStateChange({ data: 1 });
102    const held = count();
103    captured.tick();
104    captured.tick();
105    assert.equal(count(), held, "the countdown must NOT advance while the reel is
playing");
106    // reel PAUSES (state 2) ? the countdown resumes
107    captured.reel.events.onStateChange({ data: 2 });
108    captured.tick();
109    assert.equal(count(), held - 1, "the countdown resumes once the reel stops");
110  });
111

```

20. assistant (2026-06-21T14:34:41.293Z)

Now let me read the capture.html and own-model.html, the engine I18N_PLAN.md P1 section, and the SPA's LanguageSwitcher component + the deploy-config test (to understand wrangler/CF asset routing for the bundling implication).

21. user (2026-06-21T14:34:42.246Z)

```

1    <!doctype html>
2    <html lang="en">
3    <head>
4    <meta charset="utf-8" />
5    <meta name="viewport" content="width=device-width, initial-scale=1" />
6    <title>KhelSutra ? Annotate your footage & train your own model</title>
7    <meta name="description" content="With KhelSutra you can annotate your own footage and
grow a rally model that's YOURS ? on an open, MIT-licensed stack you control. The free VLC
rally-annotator turns 'watch a match' into golden labels." />
8    <meta name="robots" content="noindex" />
9    <link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg'
viewBox='0 0 32 32'%3E%3Crect width='32' height='32' rx='8' fill='%230C261F'/%3E%3Cpath d='M13
10 L23 16 L13 22 Z' fill='%2319B36B'/%3E%3C/svg%3E" />
10   <link rel="preconnect" href="https://fonts.googleapis.com" />
11   <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
12   <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;
700;800&display=swap" rel="stylesheet" />

```

```

13     <style>
14         :root{--ink:#0C261F;--green:#19B36B;--green-d:#137a4c;--lime:#C7F2B0;--paper:#F6F8F4;-
-tx:#14271F;--mut:#566B61;--line:#E1E7DF;--font:'Plus Jakarta Sans',system-ui,-apple-
system,Segoe UI,Roboto,sans-serif}
15         *{box-sizing:border-box}
16         body{margin:0;font-family:var(--font);color:var(--tx);background:#E9EDE6;line-
height:1.6;-webkit-font-smoothing:antialiased}
17         a{color:var(--green-d)}
18         .sheet{max-width:880px;margin:18px auto;background:#fff;border-
radius:16px;overflow:hidden;box-shadow:0 18px 50px -24px rgba(0,0,0,.35)}
19         .head{background:var(--ink);color:#EAF3EE;padding:24px 32px}
20         .head .top{display:flex;align-items:center;justify-content:space-
between;gap:1rem;flex-wrap:wrap}
21         .brand{display:flex;align-items:center;gap:.5rem;font-weight:800;font-
size:1.2rem;color:#fff;text-decoration:none}
22         .brand .play{display:inline-block;width:0;height:0;border-style:solid;border-width:6px
0 6px 11px;border-color:transparent transparent transparent var(--green);margin:0
.3em;transform:translateY(1px)}
23         .head h1{font-size:clamp(1.5rem,3.5vw,2.1rem);font-weight:800;letter-
spacing:-.02em;margin:1rem 0 0;max-width:20ch}
24         .head h1 .hl{color:var(--lime)}
25         .head p{color:#A7C3B7;margin:.7rem 0 0;max-width:62ch}
26         .head .nav{display:flex;gap:1rem;font-size:.9rem;margin-top:1rem;flex-wrap:wrap}
27         .head .nav a{color:var(--lime);text-decoration:none}
28         .body{padding:26px 32px 8px}
29         .sec{padding:18px 0;border-top:1px solid var(--line)}
30         .sec:first-child{border-top:0}
31         .kick{font-size:.78rem;font-weight:700;letter-spacing:.06em;text-
transform:uppercase;color:var(--green-d)}
32         .sec h2{font-size:1.4rem;font-weight:800;letter-spacing:-.01em;margin:.35rem 0 0}
33         .sec p{color:#34453d;margin:.6rem 0 0}
34         .pill{display:inline-block;font-size:.72rem;font-weight:700;border-
radius:999px;padding:.2rem .6rem;margin-left:.4rem;vertical-align:.12em}
35         .pill.live{background:rgba(25,179,107,.14);color:#0a6c45;border:1px solid
rgba(25,179,107,.35)}
36         .pill.build{background:#FAEEDA;color:#854F0B;border:1px solid #EF9F27}
37         ol,ul{margin:.6rem 0 0;padding-left:1.2rem}
38         li{margin:.25rem 0;color:#34453d}
39         li b{color:var(--tx)}
40         code{font-family:ui-monospace,SFMono-Regular,Menlo,Consolas,monospace;font-
size:.86em;background:#EEF3EC;border:1px solid var(--line);border-radius:5px;padding:.05rem
.35rem}
41         pre{background:#0C261F;color:#EAF3EE;border-radius:10px;padding:14px
16px;overflow:auto;margin:.8rem 0 0;font-size:.84rem;line-height:1.5}
42         pre code{background:none;border:0;padding:0;color:inherit}
43         .honest{display:flex;gap:.7rem;background:#fff;border:1px solid var(--line);border-
left:4px solid var(--green);border-radius:0 10px 10px 0;padding:12px 16px;margin:.9rem 0 0}
44         .honest b{color:var(--tx)}
45         .honest p{margin:.2rem 0 0;color:var(--mut);font-size:.93rem}
46         .notes{list-style:none;padding:0;margin:.8rem 0 0;display:grid;grid-template-
columns:1fr 1fr;gap:8px}
47         .notes li{display:flex;gap:.5rem;background:#FBF0ED;border:1px solid #f0d6cc;border-
radius:8px;padding:.5rem .7rem;color:#6b3a2a;font-size:.9rem;margin:0}
48         .moat{display:grid;grid-template-columns:1fr 1fr;gap:10px;margin-top:.8rem}
49         .moat div{background:#EAF6EF;border:1px solid #BfE6D2;border-radius:10px;padding:10px
12px}
50         .moat h4{margin:0;font-size:.92rem;color:var(--green-d)}
51         .moat p{margin:.2rem 0 0;font-size:.86rem;color:#34453d}
52         .shot{display:flex;align-items:center;justify-content:center;text-
align:center;background:#EEF3EC;border:1px dashed #BfD3C6;border-
radius:10px;color:var(--mut);font-size:.82rem;padding:18px;margin-top:.8rem;min-height:96px}
53         .ghbtn{display:inline-flex;align-
items:center;gap:.5rem;background:#0C261F;color:#fff;font-weight:600;font-size:.9rem;text-
decoration:none;border-radius:999px;padding:.5rem .95rem;margin-top:.9rem}
54         .cta{background:#EAF6EF;border:1px solid #BfE6D2;border-radius:12px;padding:16px

```

```

18px;margin:18px 0 6px;text-align:center}
55     .cta a{display:inline-block;background:var(--green);color:#04140D;font-
weight:700;text-decoration:none;border-radius:999px;padding:.6rem 1.3rem;margin-top:.5rem}
56     table{border-collapse:collapse;margin-top:.8rem;font-size:.86rem;width:100%}
57     th,td{border:1px solid var(--line);padding:.35rem .55rem;text-align:left}
58     th{background:#EEF3EC}
59     .foot{display:flex;justify-content:space-between;flex-wrap:wrap;gap:.5rem;padding:16px
32px 26px;color:var(--mut);font-size:.85rem}
60     .foot a{text-decoration:none}
61     @media (max-width:680px){.notes,.moat,.shots{grid-template-columns:1fr}}
62     /* annotation-tool screenshots (no footage/people ? pure UI) */
63     .shots{display:grid;grid-template-columns:1fr 1fr;gap:14px;margin-top:.9rem}
64     .shots figure{margin:0}
65     .shots img{width:100%;height:auto;display:block;border:1px solid var(--line);border-
radius:10px}
66     .shots figcaption{margin-top:.4rem;font-size:.82rem;color:var(--mut)}
67     .sample{background:#EAF6EF;border:1px solid #BfE6D2;border-radius:10px;padding:11px
14px;margin:0 0 4px;font-size:.92rem;color:#34453d}
68     .sample a{font-weight:600}
69     </style>
70     </head>
71     <body>
72     <div class="sheet">
73     <div class="head">
74     <div class="top">
75         <a class="brand" href="/">Khel<span class="play" aria-
hidden="true"></span>Sutra</a>
76         <a href="/#join" style="color:#C7F2B0;font-weight:600;text-
decoration:none">Request access ?</a>
77     </div>
78     <h1>Annotate your footage. <span class="hl">Train a model that's yours.</span></h1>
79     <p>You can't train Gemini or GPT on your court footage. KhelSutra's direction is the
inverse ? a small, <b style="color:#EAF3EE">owned</b> rally model where the moat is your
consented dataset, on a commercially-clean, inspectable stack you control.</p>
80     <div class="nav"><a href="/">? KhelSutra home</a><a href="#annotate">Annotate</a><a
href="#train">Train your own model</a><a href="#setup">Setup & use</a></div>
81     </div>
82
83     <div class="body">
84
85         <p class="sample">? <b>Want to see what KhelSutra produces?</b> A real sample reel
plays the moment you <a href="/#join">request access</a>.</p>
86
87         <section class="sec" id="annotate">
88             <div class="kick">Step 1 . real and shipping today</div>
89             <h2>Annotate your footage <span class="pill live">live today</span></h2>
90             <p><b>rally-annotator</b> is a free, open-source (MIT) <b>VLC plugin</b> that
turns "watch a match" into golden labels. While you watch your own video in VLC, you mark each
rally's start and end plus how the point ended; it writes one row per rally to a plain CSV next
to your video. <b>Your footage never leaves your machine.</b> It covers five net-separated
racquet sports (badminton, tennis, table tennis, pickleball, padel).</p>
91             <a class="ghbtn" href="https://github.com/Khelsutra/rally-annotator"><svg
width="16" height="16" viewBox="0 0 16 16" fill="currentColor" aria-hidden="true"><path d="M8
0C3.58 0 0 3.58 0 8a8 8 0 0 0 5.47 7.59c4.07.55-.17.55-.38 0-.19-.01-.82-.01-1.49-2.01.37-2.53-
.49-2.69-.94-.09-.23-.48-.82-1.13-.28-.15-.68-.52-.01-.53.63-.01 1.08.58 1.23.82.72 1.21
1.87.87 2.33.66.07-.52.28-.87.51-1.07-1.78-.2-3.64-.89-3.64-3.95
0-.87.31-1.59.82-2.15-.08-.2-.36-1.02.08-2.12 0 0 .67-.21 2.2.82a7.4 7.4 0 0 1 2-.27c.68 0
1.36.09 2 .27 1.53-1.04 2.2-.82 2.2-.82.44 1.1.16 1.92.08 2.12.51.56.82 1.27.82 2.15 0 3.07-1.87
3.75-3.65 3.95.29.25.54.73.54 1.48 0 1.07-.01 1.93-.01 2.2 0 .21.15.46.55.38A8 8 0 0 0 16
8c0-4.42-3.58-8-8-8Z"/></svg> github.com/Khelsutra/rally-annotator</a>
92             <div class="shots">
93                 <figure><figcaption>Enable it once: VLC ? View ? Rally
Annotator.</figcaption></figure>
94                 <figure><figcaption>Mark each rally's
start/end + ending reason, then Save Rally.</figcaption></figure>
95         </div>
96         <p style="margin-top:1rem"><b>The flow, per rally:</b></p>
97         <ol>
98             <li><b>Mark START</b> on the serve-contact frame (pause/scrub for
accuracy).</li>
99             <li><b>Mark END</b> when the shuttle is dead.</li>
100            <li>Pick an <b>ending reason</b>, optionally a shot count, then <b>Save
Rally</b> ? one CSV row.</li>
101            <li>Fix anything with <b>Edit / Delete / Undo</b>; re-open the same video later
to continue.</li>
102        </ol>
103        <p style="font-size:.9rem;color:var(--mut)">It's a click-driven dialog (no marking
hotkeys yet), and targets <b>VLC 3.0.x</b>.</p>
104    </section>
105
106    <section class="sec" id="train">
107        <div class="kick">Step 2 · the moat</div>
108        <h2>Train your own model <span class="pill build">open pipeline · in
build</span></h2>
109        <div class="honest" style="border-left-
color:var(--green);background:#EAF6EF;border-color:#BfE6D2">
110            <span aria-hidden="true" style="color:var(--green-d);font-size:1.2rem">?</span>
111            <div>
112                <b>You don't have to train anything to start.</b>
113                <p>KhelSutra runs on a ready-made default, so you get rally detection and
reels <b>out of the box</b> ? nothing to set up. Training on your own footage (below) is an
<b>optional</b> power-up: a model tuned to your courts, your angles, your halls.</p>
114            </div>
115        </div>
116        <p>For those who want it, your labels feed an <b>owned</b> rally model ? trained
on <b>your</b> courts, <b>your</b> angles, <b>your</b> busy multi-court halls. The pipeline is
open and inspectable:</p>
117        <pre><code>annotate (VLC) ? &lt;video>.rallies.csv ? worker train ? your
model bundle (weights + a reviewable manifest)</code></pre>
118        <div class="moat">
119            <div><h4>You own it</h4><p>A model and dataset that are yours to keep and
improve ? not rented inference.</p></div>
120            <div><h4>Commercially clean</h4><p>MIT/Apache/BSD-only stack; we never train on
a paid LLM's output.</p></div>
121            <div><h4>Private & consented</h4><p>Footage stays yours; training use is
explicit opt-in, never automatic.</p></div>
122            <div><h4>Minors protected</h4><p>Footage involving minors is never used for
training and never published.</p></div>
123        </div>
124        <div class="honest">
125            <span aria-hidden="true" style="color:var(--green-d);font-size:1.2rem">!</span>
126            <div>
127                <b>Read this honestly.</b>
128                <p>The labeling tool is real and shipping. The training pipeline is <b>open
and in active build</b> ? today it's a developer CLI; the no-GPU path trains a pipeline-
validation <b>mock</b>, and a real model needs a GPU. There is <b>no "Train now" button</b> yet,
and we make <b>no accuracy claim vs a paid LLM</b> (there's no such benchmark). The promise here
is <b>ownership, licensing, privacy and consent</b> ? not "more accurate than Gemini," and not
"one click to train."</p>
129            </div>
130        </div>
131    </section>
132
133    <section class="sec" id="setup">
134        <div class="kick">Step 3 · for power users</div>
135        <h2>Setup & use</h2>
136        <p><b>Install the plugin (VLC 3.0.x).</b> Copy <code>rally_annotator.lua</code>
into VLC's per-user Lua <code>extensions</code> folder (create it if missing), then VLC ? Tools

```

? Plugins and extensions ? <i>Reload extensions</i>, and enable it from <i>View ? Rally Annotator</i>:</p>

```

137     <pre><code>Windows:  %APPDATA%\vlc\lua\extensions\
138     macOS:    ~/Library/Application Support/org.videolan.vlc/lua/extensions/
139     Linux:    ~/.local/share/vlc/lua/extensions/</code></pre>
140     <p style="margin-top:1rem"><b>Record</b> per the <a href="/capture">one-page
capture checklist</a> (fixed tripod, full court, landscape, whole session) ? <b>annotate</b> in
VLC ? you get <code>&lt;video&gt;.rallies.csv</code>:</p>
141     <pre><code>rally_number,start_time,end_time,ending_reason,sport,shots_count
142     1,8.800,11.500,winner,badminton,9
143     2,24.389,46.589,unforced_error,badminton,21
144     3,49.183,54.683,let,badminton,</code></pre>
145     <p style="margin-top:.6rem;font-
size:.9rem;color:var(--mut)"><code>ending_reason</code> is one of: <code>unknown</code> .
<code>winner</code> . <code>forced_error</code> . <code>unforced_error</code> .
<code>service_fault</code> . <code>let</code> . <code>other</code>. Times are decimal seconds;
downstream tools read by column name.</p>
146     <p style="margin-top:1rem"><b>Train</b> (developer CLI). A GPU-free dry run
validates the whole pipeline on a CPU mock; a real model needs a GPU:</p>
147     <pre><code># GPU-free dry run (CPU mock ? pipeline validation only):
148     python -m backend.worker train --corpus-uri &lt;bucket&gt; --out-uri &lt;bucket&gt;
--version 0.1.0
149
150     # real trajectories (needs a GPU + the native WASB runner, in build):
151     python -m backend.worker train ... --trajectory-source detector --segmenter
wasb_hybrid</code></pre>
152     <p style="margin-top:.6rem;font-size:.9rem;color:var(--mut)">One join rule: keep
the video <b>byte-identical</b> to what you annotated (don't re-encode after labeling) ? labels
match videos by content hash.</p>
153     </section>
154
155     <section class="sec">
156     <div class="kick">So expectations stay honest</div>
157     <h2>What this is not</h2>
158     <ul class="notes">
159     <li>Not a one-click trainer ? training is an offline CLI today.</li>
160     <li>Not in-browser ? labeling runs in VLC on your machine.</li>
161     <li>Not an accuracy claim vs Gemini/GPT ? no such benchmark.</li>
162     <li>Shot-type, player and score detection are roadmap.</li>
163     </ul>
164     </section>
165
166     <div class="cta">
167     <b>Want in?</b>
168     <div><a href="/#join">Request alpha access</a></div>
169     </div>
170
171     </div>
172     <div class="foot">
173     <span><a href="/">khelsutra.guru</a> . <a href="https://github.com/Khelsutra/rally-
annotator">rally-annotator (MIT)</a> . Academy flyer: <a href="/flyer" hreflang="en">English</a>
. <a href="/flyer-kn" hreflang="kn" lang="kn">????</a> . <a href="/flyer-hi" hreflang="hi"
lang="hi">????</a></span>
174     <span>Reels carry a "Generated by Khelsutra.guru" mark.</span>
175     </div>
176     </div>
177     </body>
178     </html>
179

```

22. user (2026-06-21T14:34:42.670Z)

```

1      // Deploy-config guard ? the live Cloudflare deploy is git-connected (merge to `master`
2      // auto-deploys via Workers Builds). A silent drift in wrangler.toml would break the
LIVE

```

```

3      // site with NO test noticing: PR #5 was exactly this ? the wrangler `name` didn't match
4      // the connected Worker, so builds missed khelsutra.guru. These assertions pin the
deploy
5      // invariants. String-based on purpose (the site is dependency-free; no TOML parser
dep).
6      import { test } from "node:test";
7      import assert from "node:assert/strict";
8      import { readFileSync, existsSync } from "node:fs";
9      import { join } from "node:path";
10
11      const ROOT = join(import.meta.dirname, "..");
12      const toml = readFileSync(join(ROOT, "wrangler.toml"), "utf8");
13      const has = (re, msg) => assert.match(toml, re, msg);
14
15      test("wrangler `name` matches the git-connected Worker that owns khelsutra.guru", () =>
{
16          // If this drifts, Workers Builds deploys a DIFFERENT worker and the live site goes
stale (PR #5).
17          has(/^s*name\s*=\s*"hidden-king-khelsutra"/m, "wrangler name must stay 'hidden-king-
khelsutra'");
18      });
19
20      test("wrangler `main` and the assets directory point at paths that exist", () => {
21          const main = toml.match(/^s*main\s*=\s*"([^\"]+)"/m)?.[1];
22          assert.ok(main, "main = ... is missing from wrangler.toml");
23          assert.ok(existsSync(join(ROOT, main)), `main entry '${main}' does not exist`);
24
25          const dir = toml.match(/\[assets\][\s\S]*?directory\s*=\s*"([^\"]+)"/)?.[1];
26          assert.ok(dir, "[assets] directory is missing from wrangler.toml");
27          assert.ok(existsSync(join(ROOT, dir)), `assets directory '${dir}' does not exist`);
28      });
29
30      test("D1 binding is wired (the Request-Access form persists to D1)", () => {
31          has(/\[d1_databases\]\s*\s*/, "missing [[d1_databases]]");
32          has(/binding\s*=\s*"DB"/, "D1 binding must be 'DB' (code uses env.DB)");
33          has(/database_name\s*=\s*"khelsutra-access"/, "D1 database_name drift");
34          has(/database_id\s*=\s*"([0-9a-f-]{36})"/, "D1 database_id missing/!uuid ? deploy binds
nothing");
35      });
36
37      test("email-on-registration send binding + notify vars are present", () => {
38          // The Email Routing send binding is what emails the owner on each registration (PR
#8).
39          has(/\[send_email\]\s*\s*/, "missing [[send_email]] binding");
40          has(/name\s*=\s*"NOTIFY"/, "send_email binding must be named 'NOTIFY' (code uses
env.NOTIFY)");
41          has(/NOTIFY_FROM\s*=\s*"/, "NOTIFY_FROM var missing");
42          has(/NOTIFY_TO\s*=\s*"/, "NOTIFY_TO var missing");
43      });
44

```

23. assistant (2026-06-21T14:34:45.856Z)

Now let me read the engine I18N_PLAN.md P1 section and check the SPA's main.tsx / LanguageSwitcher and wrangler.toml + capture.html to fully ground the shim design and bundling implication.

24. user (2026-06-21T14:34:48.583Z)

=== wrangler.toml ===

Cloudflare Worker for khelsutra.guru ? static site + Request-Access form API.

#

One-time setup (owner, our Cloudflare account):

npx wrangler d1 create khelsutra-access **# copy the printed database_id below**
npx wrangler d1 execute khelsutra-access --remote --file=schema.sql
npx wrangler deploy **# or connect this repo via Workers Builds (git)**

```
#
name = "hidden-king-khelsutra" # the git-connected Worker that owns khelsutra.guru ? keep in
sync with Cloudflare
main = "src/index.js"
compatibility_date = "2025-06-01"

[assets]
directory = "./public"
binding = "ASSETS"

[[d1_databases]]
binding = "DB"
database_name = "khelsutra-access"
database_id = "cd02c6bf-4bd6-4a33-a333-ced86e36b511" # from `wrangler d1 create khelsutra-
access`
```

Email-on-registration: Email Routing send binding -> a VERIFIED destination address.

```
[[send_email]]
name = "NOTIFY"
destination_address = "avi.dullu@gmail.com"
```

```
[vars]
NOTIFY_FROM = "noreply@khelsutra.guru"
NOTIFY_TO = "avi.dullu@gmail.com"
=== web package.json (scripts + deps) ===
{
  "name": "khelsutra-web",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "description": "KhelSutra Studio ? operator console SPA (drives the rally indexer engine).",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview",
    "typecheck": "tsc --noEmit",
    "test": "vitest run"
  },
  "dependencies": {
    "@fontsource-variable/noto-sans-devanagari": "^5.2.8",
    "@fontsource-variable/noto-sans-kannada": "^5.2.9",
    "i18next": "^26.3.1",
    "i18next-browser-languagedetector": "^8.2.1",
    "i18next-icu": "^2.4.3",
    "react": "^19.0.0",
    "react-dom": "^19.0.0",
    "react-i18next": "^17.0.8"
  },
  "devDependencies": {
    "@tailwindcss/vite": "^4.0.0",
    "@testing-library/dom": "^10.4.1",
    "@testing-library/react": "^16.3.2",
    "@types/node": "^26.0.0",
    "@types/react": "^19.0.0",
    "@types/react-dom": "^19.0.0",
    "@vitejs/plugin-react": "^4.3.4",
    "jsdom": "^29.1.1",
    "tailwindcss": "^4.0.0",
    "typescript": "^5.7.0",
```

```

    "vite": "^6.0.0",
    "vitest": "^3.0.0"
  }
}
=== LanguageSwitcher component? ===
CompileResult.tsx
IngestPanel.test.tsx
IngestPanel.tsx
LanguageSwitcher.tsx
QueryBar.examples.test.ts
QueryBar.tsx
RallyGrid.tsx
SelectionFooter.tsx

```

25. user (2026-06-21T14:34:52.179Z)

Your configuration specifies to merge with the ref 'refs/heads/fix/dryrun-guide-font-and-ui-tracking' from the remote, but no such ref was fetched.

=== I18N_PLAN sections ===

1:# Internationalization (i18n) ? Kannada-first, by design

3:**Status:** Design (P0 deliverable; phases P1+ are owner-gated, not started).

8:> truth). P1's "static-UI retrofit" is **retargeted**: `backend/api/static` is gone, so **P1 ? the vanilla

9:> `site/` (`t()` shim) and P3 ? the `web/` React SPA (react-i18next)**, both consuming shared

11:> unchanged. Backend localization (P2) stays here in the engine repo, owner-gated.

15:## 1. Context ? why this, why now

39:**framework-agnostic**: the catalog format and locale-negotiation contract are shared, so the

49:## 2. Core architecture

51:### 2.1 The "by design" invariant ? no hardcoded user-facing strings

65:### 2.2 One canonical English source catalog, per-locale siblings

83:### 2.3 Locale-negotiation contract (shared frontend ? backend)

88:persisted user preference ? UI switcher / `?lang=<code>` ? Accept-Language header ? default "en"

94: same `lang` query param / `Accept-Language` header, so a Kannada UI gets Kannada errors.

98:### 2.4 Backend text ? keys + params, not prose

116:### 2.5 Indic font strategy ? (open decision ? see \$4)

129:### 2.6 CI enforcement (the guardrail that keeps it "by design")

141:### 2.7 Translation pipeline

158:## 3. Phased execution (owner-gated; ONE PR per slice, off `master`)

163:| **P1** | Foundation + **static-UI retrofit**: `locales/` + `en` source catalog (extract the ~40?50 UI strings ? keys), a tiny vanilla **`t(key, vars)`** shim + `data-i18n` wiring in `index.html/`app.js`, \*\*language switcher (en/kn)\*\* + `localStorage` persistence + `?lang=`, the font decision applied, and the \*\*`kn`\*\* catalog (Gemini-draft ? owner review). Reuse the existing static mount / `serve_index` in `backend/main.py`. | this | owner approval |

164:| **P2** | **Backend + report localization**: `backend/i18n/` with `tr(lang, key, params)` (same catalog format), refactor validator `message`s to keys+params, localize `HTTPException.detail` + `spec.yaml` `customer\_verdict`/`customer\_message`, wire `Accept-Language`/`?lang=`. | this | owner approval |

165:| **P3** | **SPA adoption**: **react-i18next** consuming the *same* `locales/` catalogs; documented FE?BE contract. | `avidullu/khelsutra` | needs repo access; or owner/Opus executes there |

166:| **P4** | **Conversational + coach-mode** (future): locale-aware NL query intake ? **templated localized answers** ; coach-mode recommendation **templates**. Ties to the owned-model / PLATFORM \$5A + rich-metadata work. | this + model | gated on those features existing |

177:## 4. Open decisions (for the owner)

182: validate the catalog/negotiation contract before the SPA exists._

196:## 5. Key files / seams

202:- **New:** `locales/{en,kn,hi}/`, `backend/i18n/` (`tr()` + supported-locales + negotiation),

211:## 6. Verification

217:- **Backend:** request with `Accept-Language: kn` (or `?lang=kn`) ? localized error `detail` +

227:## Guardrail alignment

26. assistant (2026-06-21T14:34:56.685Z)

The engine repo pull failed (its branch tracks a non-existent remote ref) ? I'm reading it read-only, which is fine. Let me read the key I18N_PLAN.md sections (2.1-2.3 for the negotiation contract and the data-i18n shim design, plus the P1 row context and §2.6 CI). Let me also read the LanguageSwitcher component and capture.html to ground the shim and parity.

27. user (2026-06-21T14:34:57.631Z)

```
49     ## 2. Core architecture
50
51     ### 2.1 The "by design" invariant ? no hardcoded user-facing strings
52
53     Every coach-facing string is a stable key resolved at render time against the active
54     locale.
55     Code references `t("compile.button")`, never the English prose. This is the single rule
56     that
57     makes "add a language = add a catalog" true, and it is enforced by CI (§2.6) so it
58     cannot
59     silently regress as the codebase grows.
60
61     ```
62     "Validate & Index Video"                                ? t("video.action.validate_index")
63     alert("Validation Failed! ?review the checklist") ? t("toast.validation_failed")
64     `${count} Rally Segment${count>1?'s':''} Selected` ? t("compile.selected", {count}) #
65     ICU plural
66     HTTPException(detail="Video file not found at ?")? detail = tr(lang,
67     "error.video_not_found", {path})
68     ```
69
70     ### 2.2 One canonical English source catalog, per-locale siblings
71
72     ```
73     locales/
74     en/  common.json  # source of truth ? devs author keys here
75     kn/  common.json  # Kannada (Gemini-draft ? human-review)
76     hi/  common.json  # Hindi (next; first exercise of the extensibility checklist)
77     ```
78
79     - Format: i18next-flavored JSON ? works for the vanilla shim today and
80     react-i18next in
81     the SPA, with no conversion.
82     - ICU MessageFormat for interpolation and plurals: `{count, plural, one {# rally}
83     other {# rallies}}`.
84     This matters immediately ? the current UI already hand-rolls English plurals
85     (`Rally Segment${count > 1 ? 's' : ''}`), and Kannada/Hindi plural rules differ from
86     English.
87     ICU encodes each language's rules in the catalog, not the code.
88     - `en` is the source of truth; every other locale is keyed identically (enforced by
89     the
90     key-parity check in §2.6).
91
92     ### 2.3 Locale-negotiation contract (shared frontend ? backend)
93
94     One resolution order, defined in one place and honored by both UI and backend:
95
96     ```
97
98     persisted user preference ? UI switcher / ?lang=<code> ? Accept-Language header ?
99     default "en"
100     ```
101
102     - Supported locales are enumerated in a single module (one list to grow). Anything
103     outside
104     it falls back to `en`.
105     - The backend localizes its own responses (`HTTPException.detail`, report `customer_`)
```

from the

94 same `lang` query param / `Accept-Language` header, so a Kannada UI gets Kannada errors.

95 - ****Always-fallback to `en`**** for any missing key ? a missing translation must degrade to

96 English, never to a blank or a raw key in front of a user.

97

98 **### 2.4 Backend text ? keys + params, not prose**

99

100 Validator `message`s and API errors today bake in English prose ***and*** interpolated values

101 (``f"?sharpness score of {avg} is below ? threshold of {threshold}."``). The design separates

102 the two:

103

104 - The validator/endpoint emits a ****stable code + a params dict****

(``("blur.below_threshold",`

105 `{ "score": avg, "threshold": threshold })``).

106 - The ****prose lives in the catalog****, looked up at the API boundary in the negotiated locale.

107 - ``ValidationResult`` keeps an English ``message`` for logs/back-compat, but gains an optional

108 ``message_key` + `message_params``; the report/serializer prefers the key when present. This

109 keeps internal/technical messages English (good for debugging) while customer-facing text

110 localizes.

111

112 ``spec.yaml`` ``customer_verdict`` and per-rule ``customer_message`` become ****keyed**** (or per-locale

113 overlays keyed by rule ``code``); the technical ``message`` blocks stay English (they cite

114 ``README#Qn`` and are for the operator, not the end user).

115

116 **### 2.5 Indic font strategy ? (open decision ? see §4)**

117

118 The bundled UI/watermark fonts (Outfit, Roboto Slab) ****do not cover Kannada or Devanagari****.

119 ****Noto Sans Kannada / Noto Sans Devanagari are OFL-licensed****, which is the same tension we hit

120 with the watermark font: OFL is not in the strict ****MIT/Apache/BSD**** code/data/weights guardrail.

121 Fonts are ***media assets***, not code or model weights, and OFL is explicitly designed for

122 unrestricted redistribution ? so the cleanest path is to ****carve out an explicit OFL-for-fonts**

123 **exception****, the same way we'd treat any bundled media. Options put to the owner in §4.

124

125 Note the ****video watermark already takes a `font\_path`**** (``WatermarkConfig.font_path``, ``backend/pipeline/stitcher/compiler.py``), so stamping a Kannada/Hindi watermark is a

126 ***configuration*** change (point it at a Noto font) once the font decision is made ? no new code.

127

128 **### 2.6 CI enforcement (the guardrail that keeps it "by design")**

129

130

131 Two checks, both pure-Python and runnable in this repo (no GPU/torch needed):

132

133 1. ****Key-parity test**** ? every locale catalog has ***exactly*** the ``en`` key set; fail on missing

134 keys, extra keys, or untranslated placeholder markers (e.g. a ``"__MISSING__"`` sentinel left

135 by the draft script). This is also the ****extensibility proof****: drop a stub ``ta.json`` with

136 one key and CI immediately lists every gap.

137 2. ****No-bare-strings lint**** ? a check that flags user-facing string literals not routed through

138 ``t()`` / the backend ``tr()`` (scoped to the UI handlers and the API/validator/report

```

text
139     paths). Starts as a curated allowlist so it can land without boiling the ocean.
140
141     ### 2.7 Translation pipeline
142
143     ```
144     dev authors en keys ? scripts/i18n_draft.py (Gemini, INFERENCE only) ? kn/hi draft
145                          ? native-speaker review (owner reviews first draft)
146                          ? CI key-parity + placeholder check ? merge
147     ```
148
149     `scripts/i18n_draft.py` reads `locales/en/*.json`, asks Gemini for each missing key in
the
150     target locale with the key name and any ICU placeholders preserved, and writes a
draft
151     catalog with every machine-drafted value flagged for review. It never touches `en`,
never
152     invents keys, and is a *drafting aid* ? a human reviews before merge. (Reuses the
existing
153     Gemini provider credentials/config; this is the same "paid LLM = inference, never
training"
154     posture as the rest of the system.)
155
156     ---
157
158     ## 3. Phased execution (owner-gated; ONE PR per slice, off `master`)
159
160     | Phase | Scope | Repo | Gate |
161     |---|---|---|---|
162     | P0 | This design doc (`docs/I18N_PLAN.md`) | this | the immediate
deliverable_ |
163     | P1 | Foundation + static-UI retrofit: `locales/` + `en` source catalog
(extract the ~40?50 UI strings ? keys), a tiny vanilla t(key, vars) shim + `data-i18n`
wiring in `index.html`/`app.js`, language switcher (en/kn) + `localStorage` persistence +
`?lang=`, the font decision applied, and the kn catalog (Gemini-draft ? owner review).
Reuse the existing static mount / `serve_index` in `backend/main.py`. | this | owner approval |
164     | P2 | Backend + report localization: `backend/i18n/` with `tr(lang, key,
params)` (same catalog format), refactor validator `message`s to keys+params, localize
`HTTPException.detail` + `spec.yaml` `customer_verdict`/`customer_message`, wire `Accept-
Language`/`?lang=`. | this | owner approval |
165     | P3 | SPA adoption: react-i18next consuming the same `locales/` catalogs;
documented FE?BE contract. | `avidullu/khelsutra` | needs repo access; or owner/Opus executes
there |
166     | P4 | Conversational + coach-mode (future): locale-aware NL query intake ?
templated localized answers; coach-mode recommendation templates. Ties to the owned-
model / PLATFORM $5A + rich-metadata work. | this + model | gated on those features existing |
167
168     Extensibility checklist (this is the by-design contract ? runs the same every
time):
169     > add `locales/<code>/common.json` ? `scripts/i18n_draft.py --locale <code>` ? human
review ?
170     > add `<code>` to the supported-locales list ? confirm font coverage ? CI key-parity
green.
171
172     Hindi (`hi`) is the first real exercise of this checklist ? if adding `hi` requires
touching
173     anything other than the steps above, the design failed and we fix the design, not paper
over it.
174
175     ---
176
177     ## 4. Open decisions (for the owner)
178

```

```

1      import { useTranslation } from "react-i18next";
2      import { SUPPORTED_LOCALES, DEFAULT_LOCALE, type Locale } from "../i18n";
3
4      // Each locale's label is its own autonym, shown in its own script. Kannada renders
crisply once the
5      // bundled Noto Sans Kannada font lands (tracker B4); most devices already ship a system
Kannada font.
6      const LANG_LABEL_KEY: Record<Locale, string> = {
7          en: "lang.en",
8          kn: "lang.kn",
9          hi: "lang.hi",
10     };
11
12     /**
13      * A persistent language switcher. Selecting a locale calls i18next's changeLanguage,
which the
14      * detector caches to localStorage (LANG_STORAGE_KEY) ? so the choice is the top-
priority signal on
15      * the next visit (negotiation: persisted pref wins). Icon-free + non-color active state
via
16      * aria-pressed keeps it usable for non-tech-savvy users (FRIENDLY_AND_MULTILINGUAL §4).
17      */
18     export function LanguageSwitcher() {
19         const { t, i18n } = useTranslation();
20         // Highlight the user's CHOSEN locale (i18n.language), not the resolved-for-content
one:
21         // resolvedLanguage falls back to `en` when a catalog is missing, so picking Kannada
(content
22         // still English until the kn catalog lands, B5) must still show Kannada as selected.
Compare on
23         // the base subtag so a detected "en-IN" highlights "en".
24         const active = (i18n.language || DEFAULT_LOCALE).split("-")[0];
25
26         return (
27             <nav aria-label={t("lang.label")} className="flex items-center gap-1">
28                 {SUPPORTED_LOCALES.map((loc) => {
29                     const isActive = active === loc;
30                     return (
31                         <button
32                             key={loc}
33                             type="button"
34                             lang={loc}
35                             aria-pressed={isActive}
36                             onClick={() => void i18n.changeLanguage(loc)}
37                             className={[
38                                 "rounded-full px-3 py-1.5 text-sm font-semibold transition",
39                                 isActive ? "bg-white text-ink" : "text-white/70 hover:text-white",
40                             ].join(" ")}
41                         >
42                             {t(LANG_LABEL_KEY[loc])}
43                         </button>
44                     );
45                 })}
46             </nav>
47         );
48     }
49

```

29. assistant (2026-06-21T14:35:07.053Z)

I now have a complete picture. Let me verify the catalog-home decision detail ? whether `web/src/i18n/locales` is currently the only catalog location (so B3 needs to decide between hoisting vs importing) and how the site is dependency-free (no build step). Let me confirm there's no top-level `locales/` dir and check the §4/§7-B3 references in FRIENDLY doc are fully captured.

30. user (2026-06-21T14:35:09.679Z)

```
=== top-level locales? ===
ls: cannot access 'locales': No such file or directory
=== repo root layout ===
NEXT_STEPS.md
README.md
docs
node_modules
site
web
=== web vite config alias? ===
import { defineConfig } from "vite/config";
import react from "@vitejs/plugin-react";
import tailwindcss from "@tailwindcss/vite";

// The SPA makes SAME-ORIGIN relative `/api/*` calls. In the appliance (prod) a reverse
// proxy serves the SPA and proxies `/api` to the engine on 127.0.0.1:8000
// (docs/LOCAL_APPLIANCE_ARCHITECTURE.md). In dev, Vite proxies `/api` to the local engine
// (`python -m backend.main --server`, :8000). Override with VITE_ENGINE_ORIGIN if needed.
export default defineConfig({
  plugins: [react(), tailwindcss()],
  server: {
    proxy: {
      "/api": { target: "http://localhost:8000", changeOrigin: true },
    },
  },
  test: {
    // Default env is node (the pure lib/parser tests); the i18n render test opts into jsdom via
a    // `@vite-test-environment jsdom` docblock. Match .test.tsx too so render tests are
discovered.
    environment: "node",
    include: ["src/**/*.test.{ts,tsx}"],
    // DOM-env localStorage polyfill (jsdom under Node 26 ships none); node-env files no-op.
    setupFiles: ["src/test/setup.ts"],
  },
});
=== how web imports locales (relative) ===
13:import enCommon from "../locales/en/common.json";
14:import knCommon from "../locales/kn/common.json";
15:import hiCommon from "../locales/hi/common.json";
```

31. assistant (2026-06-21T14:35:14.132Z)

I have everything needed. The catalogs currently live only at
`web/src/i18n/locales/{en,kn,hi}/common.json` (imported via relative paths in
`web/src/i18n/index.ts`); there's no top-level `locales/` despite the doc's §4 diagram showing
one. The site is dependency-free (no build step, no bundler ? `wrangler` serves `public/` raw
and `server.js` reads files directly). This is the crux of the catalog-home decision.

Now I'll compose the B3 ticket.

32. user (2026-06-21T14:36:46.261Z)

Structured output provided successfully

33. assistant (2026-06-21T14:37:01.281Z)

I've specced the B3 ticket and returned it via StructuredOutput. Summary of what I grounded it
in and the key decisions:

****Recommendation on catalog-home (the load-bearing call):**** Option A ? hoist to a repo-root
`/locales/{en,kn,hi}/common.json` as the single source of truth, with a tiny prebuild that

copies it into `site/public/locales/`. This is forced by two verified facts:

- Catalogs today live *only* at `web/src/i18n/locales/{en,kn,hi}/common.json` (imported via relative paths in `web/src/i18n/index.ts:13-15`); there is *no* top-level `locales/` yet, despite the FRIENDLY §4 / I18N_PLAN §2.2 diagrams showing one.
- The site is runtime dependency-free with *no bundler* (`wrangler [assets]` serves `public/` raw; `server.js` PUBLIC-jails paths at `server.js:57-58`), and `web/src` is outside `public/`, so the site physically cannot serve the SPA's copy ? a copy/hoist step is mandatory. Option B (site imports from `web/src`) is rejected for exactly this reason; Option C (keep home in the SPA, just copy) is the fallback if the owner wants minimal churn.

Shim design (~70 LOC, `site/public/i18n.js`, plain ES module): *dotted-key* `t(key,vars)` over the SPA's flat catalog shape; `{var}` interpolation matching *i18next*; 2-form ICU plurals via *Intl.PluralRules* (en/kn/hi are all CLDR one/other); *data-i18n* / *data-i18n-attr* DOM walk; *switcher* copied 1:1 from *LanguageSwitcher.tsx*'s aria-pressed contract.

Negotiation parity: identical order to the SPA ? persisted pref (`khelsutra.lang`, the *same* `LANG\_STORAGE\_KEY`) ? `?lang=` ? `Accept-Language`/`navigator` ? `en`, base-subtag normalized, en-fallback for unsupported locales and missing keys.

Two concrete code gotchas I flagged: *server.js* TYPES (`server.js:26-30`) has no *.json* entry (JSON serves as octet-stream ? add `application/json`); and B4's Noto font self-host was SPA-only, so the marketing pages (which load only Plus Jakarta Sans) will tofu kn/hi until the site gets its own Noto ? surfaced as a dependency, not a blocker for the en-only scaffold.

Relevant files: `C:/Users/avidu/Projects/khelsutra/web/src/i18n/index.ts`,
`C:/Users/avidu/Projects/khelsutra/web/src/i18n/i18n.test.tsx`,
`C:/Users/avidu/Projects/khelsutra/site/server.js`,
`C:/Users/avidu/Projects/khelsutra/site/src/{server,assets,ux.dom,deploy-config}.test.js`,
`C:/Users/avidu/Projects/khelsutra/site/public/{index,capture,own-model}.html`,
`C:/Users/avidu/Projects/khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`, and
`C:/Users/avidu/Projects/badminton-highlight-indexer/docs/I18N\_PLAN.md`.